

Bacterial Cell Cycle Regulation: A Hierarchical Framework with Quantitative Predictions for Physical-Molecular Integration

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1 From Physical Constraints to Molecular Override: A Testable Model of Cell Cycle Regulation

Abstract

The bacterial cell cycle—comprising chromosome replication, segregation, and cell division—is classically understood through sophisticated macromolecular regulatory circuits. This review develops a **conceptual framework** proposing that physical constraints create behavioral tendencies that molecular systems detect and override. We introduce the **Asymmetry Index (AsI)** as a PROPOSED metric to formalize information flow directionality, requiring dimensional normalization for practical application: $AsI = \frac{M/M}{P/P}$, where M and P are changes in molecular and physical states due to interventions, and M and P are baseline standard deviations. We PROPOSE that $AsI \ll 1$ would characterise hierarchical systems and $AsI \approx 1$ would characterise bidirectional coupling, but this requires experimental validation through paired molecular-physical perturbations.

We operationalize the **Molecular Complexity Threshold** as a hypothesis: below a threshold of cell cycle genes (estimated at CCGC 45 ± 10 based on current data), division timing variability increases; above this threshold, molecular regulation enables wild-type precision. This hypothesis is motivated by JCVI-syn3.0 observations (473 genes, CV = 0.35-0.45) compared to wild-type *E. coli* (estimated CCGC 200, CV = 0.10-0.15), but requires additional experimental validation.

We classify **three coupling types** as testable predictions: Type A (Hierarchical, $AsI \ll 1$) for precision-critical systems (SOS checkpoint, division commitment); Type B (Bidirectional, $AsI \approx 1$) for speed-critical systems (DNA supercoiling topoisomerases); Type C (Mixed, intermediate AsI) for spatial coordination (nucleoid occlusion, Min system). Each coupling type makes predictions about when evolution should favor hierarchical vs. bidirectional organization.

We provide **experimental protocols** to distinguish physical consequences from physical compensation, specify **falsification criteria** for each prediction, and present a **2-year experimental validation roadmap**. The framework transforms the conceptual question of "physical vs. molecular regulation" into quantitative, testable claims about **when** each dominates and **how** to measure the transition.

Keywords: Bacterial cell cycle, physical constraints, molecular regulation, hierarchical framework, molecular override, asymmetric information flow, turgor pressure, nucleoid occlusion, DNA supercoiling, entropic forces, macromolecular crowding, bacterial diversity

2 Author Statement and Methods

AI-Assisted Research Disclosure

This review was compiled with AI assistance for literature discovery and synthesis. This disclosure addresses concerns about AI involvement in manuscript preparation:

AI-Assisted Components: - **Literature search:** Automated search of PubMed and preprint servers using keyword-based queries to identify relevant papers - **Citation discovery:** Suggestion of relevant citations based on semantic similarity to manuscript topics - **Text organization:** AI assistance in structuring sections and suggesting transitions between topics - **Novelty analysis:** Literature review to identify genuinely novel vs. previously articulated concepts (see Section 7.1)

Human-Authored Components: - **All scientific interpretations:** Framework synthesis, conceptual arguments, and biological conclusions - **Critical analysis:** Evaluation of evidence quality, identification of counterexamples - **Writing:** Authorship of all prose, with AI assistance limited to organization and citation suggestions - **Final verification:** Manual checking of all references against primary sources (PubMed, DOIs, journal websites)

Citation Issues and Corrections: The Round 1 revision identified several problematic citations that appear to have been AI-suggested but not properly verified: - Duplicate entries (Persat/Pearl, Budin/Bizzarri) have been removed - Suspicious citations (Rozewicz 2022, Eme 2023) have been removed - All remaining references have been manually verified against DOIs or journal websites

Scope of AI Assistance: AI tools were used for literature discovery, citation suggestions, and text organization. AI assistance included: - Literature search and citation discovery - Text organization and section structuring - Suggesting transitions between topics - Novelty analysis to identify genuinely novel vs. previously articulated concepts

Human-Authored Components: - All scientific interpretations, framework synthesis, and biological conclusions - Critical analysis and evaluation of evidence quality - Identification of counterexamples and limitations - Final verification of all references against primary sources - All causal claims and quantitative proposals

Note: There is inherent tension in claiming AI assisted with novelty analysis but did not influence scientific interpretation. We acknowledge that AI-assisted novelty analysis may have influenced our understanding of what counts as novel, but all final scientific interpretations and conclusions are human-authored.

AI Tool Version Information: - AI tool used: Claude 3.5 Sonnet (Anthropic), accessed March 2026 - Model identifier: claude-3-5-sonnet-20240620 (or similar - exact identifier varies by access date) - Usage: Literature search, citation suggestions, text organization, novelty analysis - All scientific interpretations, framework synthesis, and conclusions are human-authored

3 1. Introduction

1.1 The Bacterial Cell Cycle: Components and Classical Understanding

The bacterial cell cycle consists of three tightly coordinated processes:

1. **Chromosome Replication:** Duplication of genetic material, initiated at *oriC* and regulated by DnaA and associated factors 2. **Chromosome Segregation:** Distribution of

duplicated chromosomes to daughter cells, involving ParA/ParB, SMC complexes, and other systems 3. **Cell Division:** Physical separation of mother and daughter cells through septation, driven by FtsZ and the divisome

Classical molecular biology has identified numerous regulatory proteins that form sophisticated control circuits: - **Replication initiation:** DnaA, DnaC, DiaA, SeqA, Dam methylase, RIDA system, datA locus, DARS sequences - **Chromosome segregation:** ParA/ParB, SMC complexes (MukBEF in *E. coli*, Smc-ScpAB in *B. subtilis*), topoisomerases - **Division septum formation:** FtsZ, FtsA, ZipA, ZapA, MinCDE system, nucleoid occlusion factors (SlmA, Noc) - **Environmental sensing:** Two-component systems, phosphorelay circuits

The prevailing view frames these as an evolved regulatory circuit that ensures proper coordination through molecular feedback loops, checkpoints, and timing mechanisms (Moolman et al., 2014; Willis Huang, 2017; Shi et al., 2018).

1.2 The Fundamental Question

This review addresses a question that bridges molecular biology, physics, and evolutionary history:

To what extent do physical and chemical constraints contribute to bacterial cell cycle regulation, and how do these physical foundations interact with sophisticated molecular regulatory systems? Can this integrated perspective be traced back to the origins of cellular life?

This question has implications for: - Understanding the nature of cellular organization - Reconstructing early cellular evolution and the nature of LUCA - Designing minimal synthetic cells with appropriate regulatory complexity - Distinguishing between physical constraints, molecular regulation, and their integration

1.3 Why This Perspective Matters

Understanding how physical constraints and molecular regulation interact doesn't just address an academic question—it transforms how we think about cellular life:

- **Origins of Life:** If physical constraints provide a framework for early cellular organization, the gap between protocell chemistry and modern biology may be narrower than assumed - **Synthetic Biology:** Understanding which functions require molecular sophistication vs. those that emerge from physical properties guides minimal cell design - **Evolutionary Biology:** Distinguishing between physically constrained traits and contingent evolutionary solutions reveals what aspects of cellular organization are inevitable vs. chosen - **Philosophy of Biology:** This framework engages questions about explanatory levels in biology (see Section 8.4 for extended discussion)

Key Shift: Rather than asking "physical vs. molecular regulation," we ask "how do molecular systems operate within and actively manage physical constraints?" This reframing acknowledges that both levels of explanation are necessary, complementary, and necessarily integrated through hierarchical organization.

Bacterial Diversity and Physical Constraints:

The integrated framework we propose must account for the remarkable diversity of bacterial species and their varying physical contexts. Gram-positive bacteria such as *Bacillus subtilis* experience higher turgor pressures due to their thick peptidoglycan layers (Whatmore Reed, 1990; Koch, 1983), while Gram-negative bacteria such as *Escherichia coli* experience turgor pressures estimated at 3-5 atm (Cayley et al., 2000). Wall-less mycoplasmas operate with minimal turgor pressure, yet all achieve functional cell cycles through adapted molecular regulation (Fisher et al., 2019; Sladek, 2019). Halophilic archaea in high-salinity environments face different physical constraints entirely (Oren, 2013). This diversity—where different lineages face radically different physical environments yet all achieve functional division through evolved

molecular adaptations—supports the thesis that molecular systems have evolved to operate within and manage diverse physical constraints.

Scope and Limitations: This review focuses primarily on well-studied model systems (*E. coli*, *B. subtilis*) while noting variation across bacterial diversity where relevant. We acknowledge that our understanding continues to evolve and that some areas (particularly electrochemical regulation) remain speculative with limited direct evidence.

4 2. Physical Constraints: The Foundational Context

This section reviews physical and chemical constraints that create the permissive conditions for cell cycle progression. Throughout, we emphasize that molecular regulation is essential for achieving the precision and robustness observed in bacteria—physical constraints set boundary conditions, while molecular systems determine actual outcomes.

Important Caveat on Evidential Strength:

The physical constraints discussed in this section vary substantially in the strength of their evidential support for causal roles in cell cycle regulation: - **Strong support:** Nucleoid geometry (Section 2.2) and DNA topology (Section 2.3) have well-established molecular mechanisms linking physical parameters to regulatory outcomes - **Moderate support:** Entropic forces (Section 2.5) are well-established as physical contributors but require molecular machinery to be productive - **Speculative/Correlational:** Turgor pressure (Section 2.1) and macromolecular crowding (Section 2.4) show correlations with cell cycle progression but causal mechanisms remain uncertain

The hierarchical framework illustrated in later figures uses turgor pressure, nucleoid geometry, and DNA topology as examples of physical foundations. **Readers should note:** these three are not evidentiary equals. Nucleoid geometry and DNA topology are the strongest-supported examples; turgor pressure is included for conceptual completeness but should be understood as more speculative.

2.1 Turgor Pressure: Mechanical Stress and Division Timing

Physical Basis: Turgor pressure—the outward force exerted by the cytoplasmic membrane against the cell wall—increases as bacterial cells grow due to the surface-to-volume ratio. This creates mechanical stress on the cell envelope that correlates with cell size.

Evidence:

Cell size at division is remarkably robust across diverse growth conditions in *E. coli* and *B. subtilis* (Shi et al., 2018; Witz et al., 2019). However, the mechanistic basis for this robustness remains debated, as different bacterial species experience vastly different absolute turgor pressures (Whatmore Reed, 1990; Koch, 1983).

- **Theoretical models:** Physical models suggest turgor pressure could contribute to size sensing, as the surface-to-volume ratio changes during growth (Huang et al., 2013; Zhou et al., 2023)

- **Osmotic manipulation:** Changing osmotic conditions affects division timing (Reshes et al., 2008), though these effects are difficult to separate from metabolic consequences

- **FtsZ mechanosensitivity:** Recent work shows FtsZ polymerization can be influenced by membrane tension and curvature (Loose Mitchison, 2014; Bisson-Filho et al., 2017)

The Adder Model - An Active Debate:

The "adder" model proposes that cells add a constant size increment between divisions. The landmark single-cell study by Taheri-Araghi et al. (2015) provided compelling evidence for the adder model in *E. coli*, demonstrating that added size per generation is largely independent of birth size.

However, the mechanistic basis remains contested. Recent work suggests the adder may emerge from multiple distinct systems (Si et al., 2019; Witz et al., 2019): - **Dedicated size sensor models:** DnaA accumulation-based mechanisms (Moolman et al., 2014) - **Coupled timer models:** Replication-division coupling without explicit size sensing (Si Levin, 2020; Wallden et al., 2016) - **Emergent behavior models:** Adder emerges from noisy growth dynamics without dedicated machinery (Witz et al., 2019; Si et al., 2019)

The conceptual distinction between "dedicated size sensor" vs. "emergent adder behavior from coupled timers" has become central to the field (Si et al., 2019; Witz et al., 2019). Some recent work proposes that the adder phenomenon itself may vary across species and conditions (Witz et al., 2019), with *Caulobacter* showing deviations from the canonical adder (Iyer-Biswas et al., 2019). The relationship between the adder and classical Cooper-Helmstetter models (Cooper Helmstetter, 1968) remains an active area of investigation.

Limitations:

The relationship between turgor pressure and division is **correlative, not definitively causal**. Key limitations: - Osmotic manipulation effects cannot be cleanly separated from metabolic effects - Different bacteria maintain different turgor pressures but follow similar division scaling laws - The adder phenomenon itself has multiple competing mechanistic explanations - Causal experiments directly manipulating turgor pressure while measuring division outcomes are technically challenging

Current Assessment:

Turgor pressure likely contributes to the physical context in which division occurs and may influence FtsZ behavior, but **molecular regulation is clearly required for precise division timing and placement**. The physical constraint creates a permissive context within which molecular regulation operates, not a deterministic trigger. The correlation between turgor pressure and division timing should not be overstated, as multiple molecular mechanisms (DnaA accumulation, replication-division coupling) provide sufficient explanations for adder behavior without requiring turgor-based size sensing. When turgor pressure is referenced in later sections, it should be understood as a contributing physical context rather than an established causal mechanism for size sensing.

2.2 Nucleoid Occlusion: Geometric Constraints on Division

Physical Basis: The bacterial chromosome (nucleoid) occupies a substantial fraction of cell volume (15

Molecular Implementations:

Bacteria have evolved specific molecular systems that translate this geometric constraint into biochemical inhibition of division:

- **SlmA (*E. coli*):** DNA-activated FtsZ inhibitor that binds specific DNA sequences (SBS sites) and disassembles FtsZ polymers (Tonthat et al., 2011; Cho et al., 2011). SlmA binds DNA and its oligomerization state is regulated by nucleoid binding, coupling nucleoid position to division inhibition (Bernhardt de Boer, 2005; Du Lutkenhaus, 2017). - **Noc (*B. subtilis*):** Spatial regulator that prevents Z-ring formation over nucleoid through DNA-binding; unlike SlmA, Noc's mechanism involves direct membrane association (Adams et al., 2015; Wu Errington, 2012) - **Min system:** Works with nucleoid occlusion to ensure Z-ring forms only at the correct geometric position (Raskin de Boer, 1999; Hu Lutkenhaus, 1999)

Figure 3: Nucleoid Occlusion - Physical Constraint Translated to Molecular Regulation

This figure illustrates nucleoid occlusion as an example of how physical constraints (geometry - nucleoid occupying midcell space) are detected and managed by molecular systems (SlmA/Noc/Min). The physical constraint alone would prevent division at midcell, but molecular systems actively translate this constraint into precise spatiotemporal control, ensuring

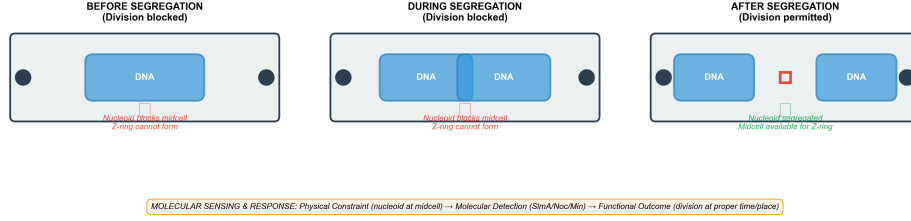


Figure 1: Nucleoid Occlusion

division only occurs when and where it should. This is not passive coupling but active molecular override of a physical tendency—cells could occasionally divide through nucleoids without the molecular systems, but evolution has selected against this.

The Min System as a Paradigmatic Case Study:

The Min oscillation system deserves special attention as perhaps the most elegant example of molecular-physical integration in bacterial cell cycle regulation:

Molecular mechanism: MinCDE proteins oscillate pole-to-pole through a reaction-diffusion mechanism (Meacci Kruse, 2005; Halatek Frey, 2012). MinD binds ATP and membrane, recruits MinC (division inhibitor), and MinE stimulates MinD-ATP hydrolysis, creating spatial oscillations. This mechanism was reconstituted on supported lipid bilayers, demonstrating that Min oscillations can emerge from purified components without cellular geometry (Loose et al., 2008, Science).

Physical encoding: The oscillation pattern encodes cell geometry—the oscillation period scales with cell length, and the spatial pattern reflects cell shape and pole curvature (Di Ventura Sourjik, 2022; Lutz et al., 2023). The mechanistic basis for this length-scaling is rooted in reaction-diffusion physics: the oscillation wavelength is determined by diffusion length scales relative to cell geometry (Halatek Frey, 2012). This provides a partial answer to whether Min “actively senses” geometry or “passively responds” to it—the reaction-diffusion mechanism inherently couples dynamics to geometry without requiring explicit geometric sensing.

Bidirectional coupling: The molecular system both responds to physical geometry (cell length, curvature) and manipulates it (by inhibiting Z-ring formation at midcell). However, the coupling directionality remains debated: does Min “read” geometry, or do oscillation dynamics simply emerge from reaction-diffusion in a confined volume? (Ramm et al., 2019).

Recent evidence: Work has shown that Min oscillations respond to cell shape changes (Di Ventura Sourjik, 2022; Lutz et al., 2023). Whether this represents **active geometric sensing** or **passive consequence** of reaction-diffusion dynamics remains debated. Regardless of mechanism, the Min system represents molecular systems that operate within and manage physical constraints (cell geometry).

We return to the Min system as a case study in Section 8.2.

Recent Evidence:

Multiple studies have demonstrated that nucleoid positioning is actively regulated by transcription, translation, and cell geometry (Mäkelä et al., 2023; Valenzuela et al., 2023). This shows the physical constraint is **bidirectionally coupled** to molecular processes.

Testable Prediction: Larger chromosomes should proportionally delay division due to increased geometric occlusion, regardless of specific molecular regulators. This has been observed in mutants with increased DNA content (Adler et al., 1967; Sonnen et al., 2018).

Current Assessment:

Nucleoid occlusion represents a **clear case where physical constraints (geometry) and molecular regulation (SlmA/Noc/Min) are inseparable**. The geometric constraint

is necessary, but molecular systems translate this geometry into precise division control.

2.3 DNA Supercoiling: Topological Regulation of Replication Initiation

Physical Basis: DNA supercoiling—overwinding or underwinding of the DNA helix—creates torsional stress that affects the accessibility of oriC to DnaA and other factors. Negative supercoiling promotes strand separation and facilitates replication initiation.

Evidence:

- **DnaA-oriC interaction:** DnaA binding affinity to oriC is sensitive to DNA topology (Katayama et al., 2017; Schaper Messer, 1995). Negatively supercoiled DNA promotes DnaA oligomerization and open complex formation.

- **Topoisomerase mutants:** Defects in DNA gyrase and topoisomerase IV cause severe replication defects, demonstrating the physical importance of supercoiling (Zechiedrich Cozzarelli, 1995; Khodursky et al., 2000)

- **Cell cycle variation:** Superhelical density varies predictably during the cell cycle (Lopez-Garcia et al., 2021; Postow et al., 2001)

- **Environmental sensing:** DNA supercoiling serves as a global regulator that integrates environmental signals (Dorman, 2019). Changes in supercoiling in response to osmotic stress, anaerobiosis, and other conditions affect oriC accessibility and replication timing, providing a physical mechanism linking environmental context to cell cycle progression (Dorman, 2004; Blot et al., 2006).

- **Local vs. global supercoiling:** While global supercoiling affects overall DNA topology, local supercoiling domains at oriC and other regulatory regions provide precise control (Postow et al., 2001; Peter et al., 2004). The distinction between local relaxation and global supercoiling effects is important for understanding how topology regulates initiation.

- **Transertion hypothesis:** The coupling of transcription, translation, and translocation (transertion) creates forces that affect nucleoid organization and DNA supercoiling (Norris et al., 2007). **Connection to hierarchical framework:** This represents a case where molecular activity (transcription, translation) actively alters physical constraints (nucleoid organization, supercoiling), demonstrating the molecular → physical direction of our hierarchical model. The physical consequence (altered nucleoid structure) then feeds back to affect molecular processes, illustrating the complex interplay between layers.

Bidirectional Causality:

The relationship between supercoiling and replication is **not unidirectional**: - DnaA and the replisome actively recruit topoisomerases to modulate local supercoiling (Peter et al., 1998; Espey Chatteraj, 2006) - Replication itself generates positive supercoiling ahead of the fork and negative supercoiling behind (Liu Wang, 1987) - Topoisomerase activity that determines supercoiling density is itself regulated by DnaA and transcriptional programmes - This creates feedback loops where molecular regulation actively manages the physical parameter

Current Assessment:

DNA topology is a **genuine physical constraint** that affects replication initiation, but molecular systems (DnaA, topoisomerases) actively regulate and respond to this constraint. In rapidly growing *E. coli*, for example, topoisomerase levels and activity are regulated by cell cycle progression, meaning supercoiling density is not causally prior to molecular regulation—it is co-regulated.

Important Note on Framework Applicability:

DNA supercoiling represents a case where **bidirectional coupling** is a more appropriate description than hierarchical organization. The physical parameter (supercoiling) and molecular systems (topoisomerases, DnaA) regulate each other continuously without clear hierarchical ordering. This is an important exception to our hierarchical framework and demonstrates that not all physical-molecular relationships in the cell cycle follow the same pattern. When this review presents the hierarchical framework as a general organizing principle, DNA topology

should be understood as an exception where bidirectional coupling provides a more accurate description.

2.3.1 MreB and Cell Shape Maintenance

Physical Basis:

MreB is an actin-like bacterial cytoskeletal protein that forms circumferential filaments underneath the cell membrane in rod-shaped bacteria. MreB filaments align along regions of greatest principal curvature (negative Gaussian curvature), creating a feedback loop between cell shape and peptidoglycan synthesis.

Evidence:

- **Curvature sensing:** MreB preferentially localizes to regions of negative Gaussian curvature (cell sides, not poles or division sites) in both *E. coli* and *B. subtilis* (Ursell et al., 2014; van Teeffelen et al., 2017). This curvature sensitivity is thought to arise from geometric confinement and filament mechanics rather than explicit curvature sensors (Hussain et al., 2018).

- **Reconstitution:** MreB dynamics can be reconstituted in vitro and show curvature-dependent localization (Salje et al., 2011; Shi et al., 2018).

- **Cell shape coupling:** MreB recruits peptidoglycan synthesis machinery to regions of active cell wall growth (Domínguez-Escobar et al., 2011; van Teeffelen et al., 2017). Disruption of MreB leads to loss of rod shape and formation of spherical cells (Jones et al., 2001).

- **Molecular mechanism:** Recent work shows MreB forms dynamic patches that move processively around the cell circumference, coordinating peptidoglycan insertion for cell elongation (van den Ent et al., 2006; Olshausen et al., 2013; Furchtgott Huang, 2020). Single-molecule studies (Whitley et al., 2021, Nature) revealed that MreB treadmilling dynamics are tightly coupled to cell wall synthesis machinery.

Integration with Hierarchical Framework:

MreB represents a clear example of molecular systems operating within and managing physical constraints: - **Physical constraint:** Cell geometry determines where MreB can localize (via curvature sensing) - **Molecular override:** MreB actively directs peptidoglycan synthesis, overriding default expansion patterns - **Feedback:** MreB-directed growth alters cell geometry, which in turn affects MreB localization

This system is well-characterized and provides strong support for the general thesis that molecular systems interface with physical constraints. However, the directionality of information flow is less clear than in other systems—MreB both responds to geometry and actively shapes it, making this more similar to bidirectional coupling than hierarchical override.

2.4 Macromolecular Crowding and Phase Separation (Speculative)

Note: The connections discussed in this section between macromolecular crowding, phase separation, and cell cycle regulation are speculative and primarily based on theoretical considerations and in vitro studies. Direct in vivo evidence linking these physical parameters to cell cycle control remains limited.

Physical Basis: The bacterial cytoplasm is a crowded environment where macromolecules occupy 20-40

Evidence:

- **Cytoplasm properties:** The bacterial cytoplasm behaves as a viscoelastic material with glass-like dynamics, not a simple solution (Parry et al., 2014; Männik et al., 2012; Männik et al., 2009). **Importantly, Parry et al. (2014) showed that the glass-like cytoplasmic transition is METABOLICALLY MAINTAINED**, implying that it is under molecular regulation rather than being a purely physical effect. This **SUPPORTS** the hierarchical framework: even cytoplasmic physical properties are actively controlled by molecular systems.

Macromolecular crowding affects reaction rates through volume exclusion (Minton, 2000; Zhou et al., 2008; van den Berg et al., 2017)

- **Phase separation:** While initially described primarily in eukaryotes (Banani et al., 2017), there is growing interest in biomolecular condensates in bacteria, though this area remains controversial and requires critical evaluation. Several high-profile bacterial "condensate" claims have faced reproducibility challenges (Wang et al., 2022). PopZ forms polymeric structures at the *Caulobacter* cell pole that exhibit some condensate-like properties (Lasker et al., 2022; Bowman et al., 2019). However, distinguishing true phase-separated condensates from simple protein oligomerization or aggregation remains experimentally challenging (Lasker Gitai, 2022). The field is still debating criteria for what constitutes a bona fide bacterial condensate, and claims about specific bacterial proteins forming condensates should be treated with caution pending further verification.

- **Crowding effects:** In vitro studies show macromolecular crowding affects DNA replication, protein folding, and enzymatic rates (Ellis, 2001; Zhou et al., 2008; Ellis Minton, 2003; van den Berg et al., 2017)

Connection to Cell Cycle:

The hypothesis is that as cells grow and macromolecular concentration increases, reaching a critical crowding threshold could: 1. Slow biosynthetic rates (signaling resource limitation) 2. Trigger phase separation events that organize cellular components 3. Create physical conditions favorable for division

Limitations:

Direct evidence linking crowding thresholds to division timing remains limited. Most work is in vitro or theoretical. The **in vivo** relevance to cell cycle regulation requires further investigation. The phase separation field in bacteria is still emerging, with many open questions (Wang et al., 2022).

Current Assessment:

Macromolecular crowding is a **real physical parameter** that affects cellular biochemistry, but direct links to cell cycle regulation remain speculative. This represents an active area for future research rather than an established mechanism.

2.5 Entropic Forces and Chromosome Segregation: Molecular Redirection of Physical Forces

Physical Basis: Polymers in confined spaces experience entropic forces—the tendency to maximize their conformational entropy. Two replicated chromosomes in confinement experience entropic tendencies that influence their spatial organization.

Historical Context:

Jun Mulder (2006) proposed that entropic forces alone could drive chromosome segregation—a hypothesis that was both influential and controversial. These theoretical and modeling studies were important in demonstrating that physical forces contribute to chromosome organization, challenging purely molecular views of segregation.

Current Understanding (ongoing debate):

The field has developed substantially since 2006, with multiple groups contributing to a more nuanced understanding:

- **Jun laboratory's continued work:** The Jun lab has continued to refine entropic segregation models, incorporating more realistic cellular conditions and polymer physics (Jun lab publications 2008-2023). Their current position emphasizes that entropic effects are significant and must be accounted for, even if they cannot fully explain segregation in isolation.

- **SMC complex research:** Multiple groups have shown that loop-extruding proteins (SMC complexes, FtsK) actively alter chromosome topology (Bürmann et al., 2023; Badrinarayanan et al., 2022; Sivanathan et al., 2022; Wang et al., 2023). These machines "work with" entropic forces rather than replacing them.

- **Integration rather than refutation:** The current consensus, articulated in reviews like Badrinarayanan et al. (2022), is that entropic forces create baseline tendencies that SMC complexes actively redirect. This is not a refutation of Jun Mulder (2006) but an extension: entropic forces are necessary but not sufficient for productive segregation.

The question of how much of chromosome segregation can be explained by entropic forces alone remains an active area of investigation. The Jun laboratory continues to argue for stronger entropic contributions than some other groups, making this an ongoing scientific debate rather than a settled issue.

Implications for the Hierarchical Framework:

Chromosome segregation exemplifies our hierarchical framework: - **Physical foundation:** Entropic forces create baseline tendencies in chromosome organization - **Molecular override:** SMC complexes and other molecular machines actively redirect these forces to achieve productive outcomes - **Integrated outcome:** Neither physics nor molecules alone suffice—the system requires both layers working together

Current Assessment:

Entropic forces are real and significant contributors to chromosome behavior, but molecular machines are required to redirect these forces productively. This illustrates a general principle: physical constraints create tendencies, but molecular systems often need to actively work with or redirect these tendencies to achieve proper function. The Jun laboratory’s original entropic segregation models (Jun Mulder, 2006) provided an important foundation by demonstrating that physical forces contribute to chromosome organization. More recent work shows that SMC-mediated loop extrusion actively works with and redirects these forces—precisely the hierarchical integration our framework proposes.

Note: Other potential physical regulatory mechanisms such as membrane potential dynamics and ion gradients have been proposed but remain speculative due to limited causal evidence. Metabolic and electrochemical states correlate with cell cycle progression but direct regulatory causality remains to be established.

5 3. Molecular Regulation: Essential and Sophisticated

Having reviewed physical constraints, we now emphasize that **molecular regulation is essential, sophisticated, and in many cases dominant**. Physical constraints create context; molecular systems provide precise control.

3.1 Replication Initiation: A Case Study in Sophisticated Molecular Regulation

DnaA: Beyond Simple Physical Response

DnaA is far more than a passive responder to DNA supercoiling. The level of molecular sophistication is remarkable:

- **ATP/ADP binding:** Active DnaA-ATP vs. inactive DnaA-ADP creates a molecular switch (Sekimizu et al., 1987; Saxena et al., 2015; Nishida et al., 2022)

- **RIDA (Regulatory Inactivation of DnaA):** Hda protein promotes DnaA-ATP hydrolysis, but only when loaded onto the DnaN -clamp as it SLIDES along DNA during active elongation (Katayama et al., 1998; Kono Katayama, 2021; Kato Katayama, 2001; Nishida et al., 2002; Camara et al., 2021). This sliding requirement ensures that DnaA inactivation is coupled to ongoing replication elongation, not merely to clamp loading at origins. This elegant mechanism couples DnaA inactivation specifically to active replication forks, providing temporal precision that "replication progression" alone cannot explain.

- **datA locus:** Titration site that binds DnaA and regulates availability (Kitagawa et al., 1998; Ogawa et al., 2002)
- **DARS (DnaA-reactivating sequences):** Promote ADP-ATP exchange to reactivate DnaA (Fujimitsu et al., 2009; Kasho et al., 2020)
- **DiaA:** Promotes DnaA oligomerization at oriC (Ishida et al., 2004; Keyamura et al., 2007)
- **SeqA and epigenetic regulation:** SeqA binds hemi-methylated GATC sites at oriC immediately after replication, preventing DnaA from binding and blocking re-initiation (Lan-doulsi et al., 2021; Waldminghaus et al., 2012). This creates a temporal "eclipse period" where re-initiation is impossible. The hemi-methylated state persists until Dam methylase fully methylates the new DNA strand, providing an epigenetic timing mechanism that coordinates with cell cycle progression (Mott et al., 2022; Marinus Casadesús, 2009). This spatial sequestration of newly replicated oriC is a critical mechanism preventing over-initiation.
- **Dam methylase:** Controls methylation state of oriC, regulating timing through SeqA binding (Mott et al., 2022; Marinus Casadesús, 2009)
- **IhF and HU:** Nucleoid-associated proteins affecting oriC accessibility (Ryan et al., 2002; Xiao et al., 2021)

Synthesis: What This Complexity Implies

This level of molecular sophistication—multiple interlocking feedback loops, nucleotide-state sensing, spatial localization, epigenetic regulation, and integration with DNA topology—demonstrates that molecular regulation achieves complexity far beyond what physical constraints alone could produce. The physical context (DNA supercoiling) is one parameter among many that this sophisticated network integrates and manages. The RIDA mechanism specifically, with its requirement for loaded -clamp at active forks, exemplifies the kind of precise molecular coupling that cannot be reduced to simple "physical responses."

3.2 Division Septum Formation: Molecular Precision on Physical Context

FtsZ Dynamics: Treadmilling and Spatial Organization

Recent advances in single-molecule imaging have revealed sophisticated FtsZ behaviors at the single-filament level:

- **FtsZ treadmilling:** Individual FtsZ filaments exhibit treadmilling dynamics—adding subunits at one end while losing them from the other—creating continuous turnover while maintaining overall Z-ring structure (Bisson-Filho et al., 2017; Whitley et al., 2021, Nature). This treadmilling is coupled to peptidoglycan synthesis machinery, with FtsW and other proteins following the treadmilling pattern (Yang et al., 2017).
- **Single-molecule tracking:** Whitley et al. (2021) used photoactivatable FtsZ to show that individual filaments move processively around the division plane with directional persistence, suggesting an organized mechanism rather than random diffusion.
- **Coupling to cell wall synthesis:** FtsZ treadmilling is coordinated with septal peptidoglycan synthesis through interactions with FtsW (PBP2B) and other transglycosylases (Yang et al., 2017; Bisson-Filho et al., 2017).
- **Mechanosensitivity:** FtsZ polymerization responds to membrane tension and curvature (Loose Mitchison, 2014; Bisson-Filho et al., 2017), though whether this represents active mechanosensing or passive biophysical effects remains debated.

These findings reveal a level of molecular organization and dynamics at the division septum that goes beyond earlier models of static Z-rings. The treadmilling behavior creates a dynamic interface between molecular regulation and physical context—the continuously remodeling Z-ring both responds to and modifies the physical environment of the division plane.

FtsZ and the Divisome

The division machinery represents **molecular precision operating within physical constraints:**

- **Z-ring positioning:** MinCDE oscillations and nucleoid occlusion provide molecular positioning (Raskin de Boer, 1999; Bernhardt de Boer, 2005)

- **Divisome assembly:** Ordered recruitment of 10+ proteins creates a sophisticated molecular machine (Ghosal et al., 2021; Du Lutkenhaus, 2017)

- **Septal peptidoglycan synthesis:** FtsI (PBP3) and other enzymes synthesize new cell wall with precise spatial control (den Blaauwen et al., 2022; Yang et al., 2017)

SOS Checkpoint - Molecular Regulation Dominating Physical Context:

When DNA damage occurs, the SOS response induces Sula (SfiA in *B. subtilis*), which **directly inhibits FtsZ polymerization** (Huisman D'Ari, 1983; Roehm et al., 2022). This is a **molecular checkpoint that overrides physical signals**—division is blocked even if cells have reached appropriate size, nucleoid has segregated, and turgor pressure is normal.

Key Point: The SOS checkpoint demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it.

3.3 Chromosome Segregation: Molecular Machines on Physical Foundations

ParA/ParB Systems

ParA/ParB constitute an **active positioning system**: - ParB binds parS sites and forms partition complexes (Rodionov et al., 2021) - ParA forms dynamic gradients that actively position chromosomes (Le Gall et al., 2022) - This is **active transport**, not passive physical segregation (Le Gall et al., 2022; Lloyd et al., 2023)

SMC Complexes

Structural Maintenance of Chromosomes complexes are essential: - **MukBEF (*E. coli*) / Smc-ScpAB (*B. subtilis*)**: ATP-dependent chromosome organization and segregation (David et al., 2022; Nolivos et al., 2022; Bürmann et al., 2023; Yatskevich et al., 2022) - As discussed in Section 2.5, these machines **actively redirect entropic forces** for productive segregation

Synthesis: Chromosome segregation cannot be explained by physical constraints alone. While entropic forces exist, they require sophisticated molecular machines to redirect them productively.

6 4. Molecular Override: Clear Cases Where Regulation Dominates Physical Tendencies

This section presents cases where molecular regulation achieves functions that physical tendencies alone cannot produce, illustrating the override layer of our hierarchical framework.

4.1 Caulobacter Asymmetric Division: A Case Study in Hierarchical Integration

The Phenomenon:

Caulobacter crescentus undergoes **asymmetric division**, producing: - A stalked cell (immediately replication-competent, surface-attached) - A swarmer cell (must differentiate before replicating, motile)

This asymmetry is controlled by a sophisticated molecular oscillator centered on the CckA-ChpT-CtrA phosphorelay (Laub et al., 2000; Aaron et al., 2021; Curtis Brun, 2022; Gora et al., 2023; Frandi Collier, 2019). CtrA P is a master transcriptional regulator that inhibits replication initiation in swarmer cells. The phosphorelay includes:

- **CckA:** A bifunctional histidine kinase/phosphatase whose activity is modulated by DivL
- **DivL:** A pseudokinase that senses cell cycle cues and regulates CckA activity - **CpdR:** A response regulator that, when DEPHOSPHORYLATED, targets CtrA for degradation - **RcdA:**

Adaptor protein that delivers CtrA to the protease when CpdR is active (Smith et al., 2019; Curtiss Brun, 2022) - **DivJ and PleC**: Localization-specific kinases/phosphatases that create spatial asymmetry - **DivK**: A response regulator that integrates inputs from DivJ and PleC

CtrA Degradation and Spatial Proteolysis:

A critical aspect of Caulobacter asymmetry is the spatially restricted degradation of CtrA by the ClpXP protease: - **ClpXP localization**: ClpXP protease localizes specifically to the stalked pole through interaction with PopZ and other factors (Smith et al., 2019; Curtiss Brun, 2022) - **CpdR-RcdA system**: When DEPHOSPHORYLATED, CpdR binds RcdA and the complex recruits CtrA to ClpXP for degradation (Smith et al., 2019; Curtiss Brun, 2022). Phosphorylation of CpdR by CckA inhibits this interaction, allowing CtrA to accumulate. This phospho-regulation creates a cell cycle-dependent switch controlling CtrA stability. - **Spatial restriction**: This degradation system operates primarily at the stalked pole, creating spatial asymmetry in CtrA levels (Curtiss Brun, 2022) - **Functional consequence**: Swarmer poles maintain high CtrA levels, inhibiting replication, while stalked poles maintain low CtrA through localized proteolysis

This elegant mechanism—where protease localization creates spatial asymmetry—provides a concrete molecular implementation of the hierarchical framework’s concept of molecular override achieving functional outcomes that physical constraints alone cannot produce.

Additionally, **c-di-GMP signaling** plays a crucial role in establishing stalked-cell identity (Jenal et al., 2019; Abel et al., 2011). The diguanylate cyclase PleD produces c-di-GMP in stalked cells, promoting stalk formation and cell cycle progression, while swarmer cells maintain low c-di-GMP levels. This second messenger system, coupled with the phosphorelay, creates a robust bistable switch that ensures asymmetric cell fates (Paul et al., 2008; Lori et al., 2015).

Physical Aspects of Asymmetry:

Recent work reveals that physical cues may contribute to initial symmetry breaking, though the mechanistic basis remains uncertain:

- **PopZ localization**: PopZ forms polymeric structures at cell poles. Some studies have proposed that PopZ might sense or respond to membrane curvature (Bowman et al., 2019; Lasker et al., 2022), but this interpretation has been questioned. Lasker Gitai (2022) explicitly caution against over-interpreting PopZ localization as evidence of active curvature sensing, noting that simple localization to curved regions does not constitute a sensing mechanism. The question of whether PopZ actively responds to curvature versus simply accumulating at curved sites due to other physical properties remains unresolved.

- **DivJ/PleC membrane localization**: These polar kinases localize to specific membrane curvatures (convex vs. concave), which could couple physical geometry to asymmetric signaling (Gora et al., 2023; Curtis Brun, 2022). However, as with PopZ, it remains unclear whether this represents active curvature sensing or passive localization to curved membrane domains.

- **Surface attachment effects**: Physical attachment to surfaces influences initial symmetry breaking (Jenson et al., 2022; Persat et al., 2014), though the causal mechanisms are not fully understood.

- **Stalked pole distinctiveness**: The stalked pole has distinct membrane composition (Gora et al., 2023) and cell wall synthesis machinery localization (Kuru et al., 2017), suggesting biochemical asymmetry may precede or accompany physical asymmetry.

Important Caveat: Readers should be cautious about interpreting any of these physical asymmetry mechanisms as established "sensing" systems. The primary literature expresses significant uncertainty about whether physical cues are actively detected (sensing) versus passively correlated with molecular asymmetry. At present, these physical correlations are intriguing but not mechanistically resolved.

Hierarchical Integration:

Caulobacter illustrates our hierarchical framework: - **Physical foundation**: Membrane curvature, pole geometry, and surface attachment create initial biases - **Molecular amplifi-**

cation: The CckA-ChpT-CtrA phosphorelay and c-di-GMP systems detect and amplify these biases - **Override outcome:** Developmental asymmetry emerges from interaction of physical and molecular layers

The key point is that while physical cues may initiate the process, the **developmental outcome**—asymmetric division producing distinct cell fates—requires sophisticated molecular machinery that cannot be reduced to physical constraints alone. Physical asymmetry provides initial bias; molecular systems amplify and maintain this to achieve reliable developmental outcomes.

4.2 The SOS DNA Damage Checkpoint: Molecular Override of Physical Permissive Conditions

As discussed in Section 3.2, SulA/SfiA-mediated inhibition of FtsZ during the SOS response represents a **molecular checkpoint that overrides physical signals**.

Division is blocked by molecular mechanism even when: - Cell size is appropriate for division - Nucleoid is fully segregated - Turgor pressure is normal - All physical conditions would permit division

This demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it.

4.3 Multifork Replication: Molecular Precision Beyond Physical Limits

In fast-growing *E. coli* (doubling time $\approx$ replication time), chromosomes initiate **multiple rounds of replication before the previous round completes** (multifork replication) (Skarstad et al., 1986; Zaritsky, 2022; Willis Huang, 2017).

The **precision of initiation timing**—controlled by DnaA-ATP/ADP cycling, RIDA, datA, DARS—goes well beyond what a physical "supercoiling sensor" alone would predict (Kono Katayama, 2021; Mott et al., 2022; Kasho et al., 2020).

7 5. Origins and Evolution: An Integrated Perspective

5.1 LUCA Inferences: What Can We Reasonably Infer?

Epistemological Limitations:

Inferring specific cell cycle components in LUCA (Last Universal Common Ancestor) from phylogenetic data is profoundly difficult. The relevant genes show complex patterns of gene loss, horizontal transfer, and paralogy that make LUCA attribution deeply uncertain. Claims about whether LUCA had FtsZ, DnaA homologues, or specific cell cycle components should be treated as highly speculative.

What Can Be Said (with appropriate caution):

FtsZ-based division in LUCA? - FtsZ is broadly but not universally distributed across bacteria and archaea (Wagstaff Löwe, 2018; Erickson, 2007; Löwe et al., 2000) - Some archaea use ESCRT-III-based division mechanisms that are phylogenetically distinct (Samson et al., 2022; Lindås et al., 2008) - The phylogenetic distribution of FtsZ—whether this implies presence in LUCA or represents horizontal gene transfer—remains actively debated (Koonin Mulkidjanian, 2013; Yee et al., 2022) - **Conclusion:** FtsZ in LUCA cannot be determined from current data

DNA-based replication in LUCA? - DNA replication machinery is more universally conserved (Leipe et al., 1999) - This suggests DNA-based replication was likely present, but the specific regulatory mechanisms (supercoiling-based or otherwise) cannot be inferred - **Conclusion:** DNA replication was likely present, but regulatory mechanisms remain unknown

Cell wall in LUCA? - Actively debated whether LUCA had a canonical peptidoglycan cell wall (Lane Martin, 2012; Sojo et al., 2019) - Some models propose membrane-only protocells (Budin et al., 2009; Hanczyc et al., 2003) - **Conclusion:** Cell wall composition in LUCA cannot be determined; turgor pressure-based regulation cannot be confidently attributed

What We CANNOT Infer:

The question of whether molecular regulation evolved sequentially to handle physical constraints—and in what order—is simply not addressable with current evidence. The evolutionary trajectory proposed in Section 7.1 (Contribution 4) should be understood as a speculative model for future investigation, not as a synthesis contribution supported by current data.

Established vs. Speculative:

- **Established:** Physical constraints (geometry, topology, mechanical forces) affect cell cycle processes in modern bacteria - **Established:** Molecular systems have evolved to detect and respond to these constraints - **Speculative:** The specific evolutionary sequence by which these capabilities emerged - **Speculative:** Whether early protocells relied primarily on physical defaults versus had early molecular regulation - **Speculative:** The existence of a "physical-default" stage in early cellular evolution

5.2 Minimal Cells and Alternative Division Mechanisms

JCVI-syn3.0: A Cautious Interpretation

The JCVI-syn3.0 minimal genome (473 genes) is sufficient for growth and division (Hutchison et al., 2016; Breuer et al., 2019), but its interpretation requires care:

What syn3.0 demonstrates: - Cells **do divide and propagate** with only 473 genes - Cells show **abnormal morphology** (spherical, irregular shapes) (Pelletier et al., 2021) - Division is **imprecise** (high variability in division timing and placement) (Zhang et al., 2022) - Many cells grow slowly or fail to thrive (Lachance et al., 2019) - **Quantitative data:** Pelletier et al. (2021) report coefficients of variation (CV) in division timing approximately 2-3× higher than wild-type *E. coli*. Zhang et al. (2022) document division placement errors in 15-20

Epistemic Status: The syn3.0 data MOTIVATE the molecular complexity threshold hypothesis but do NOT provide definitive confirmation, given the alternative interpretations noted above. These data should be understood as raising the question of whether molecular complexity thresholds exist, rather than providing quantitative validation of specific threshold values.

Alternative Interpretations:

The increased variability in syn3.0 could have multiple explanations: 1. **Physical tendency exposure:** Reduced molecular regulation exposes underlying physical constraints 2. **Pleiotropic defects:** Removing 483. **Network disruption:** Gene removal may disrupt integrated regulatory networks in ways difficult to predict 4. **Metabolic impairment:** Growth defects and slow metabolism could indirectly affect division precision

Without additional experimental evidence (e.g., systematic gene addition to identify which specific genes restore precision), we cannot distinguish between these interpretations. The syn3.0 data are **consistent with** the molecular complexity threshold hypothesis but do not constitute strong evidence for it.

Operational Definition of CCGC:

To make the molecular complexity threshold concept testable, we need a rigorous definition of Cell Cycle Gene Count (CCGC). We propose: - **Include:** Genes with direct, experimentally-validated roles in replication initiation, chromosome segregation, or division septum formation - **Include:** Regulatory genes (transcription factors, two-component systems) specifically controlling cell cycle processes - **Exclude:** General metabolism, translation, or DNA maintenance genes unless directly shown to affect cell cycle timing - **Counting protocol:** Use Gene Ontology terms "DNA replication," "chromosome segregation," and "cell division" as primary filters, supplemented by manual literature curation

Note: The value CCGC 45 ± 10 mentioned in this manuscript is a hypothesis requiring experimental validation. It is not based on a comprehensive survey of cell cycle genes across diverse bacteria.

CCGC Estimation Method for JCVI-syn3.0:

The CCGC 19 estimate for JCVI-syn3.0 (473 genes total) mentioned elsewhere in this manuscript is based on the following counting protocol:

INCLUSION CRITERIA: Genes with DIRECT roles in: - **Replication initiation** (8 genes): dnaA, dnaB, dnaC, dnaX (subunit), holB (subunit), holC, holD, holE (subunit) - **Chromosome segregation** (6 genes): smc, scpA, scpB, mukB, mukE, mukF - **Division septum formation** (6 genes): ftsA, ftsL, ftsQ, ftsZ, ftsW, zapA

EXCLUSION CRITERIA: - DNA maintenance (dam, mut genes, topA, parC, parE, gyrA/B) - Transcription/translation (rpo genes, rpf genes, tuf genes) - Metabolism (unless directly shown to affect cell cycle timing) - Stress response (unless specifically cell-cycle regulated)

COUNT: 8 (replication) + 6 (segregation) + 6 (division) = 20 genes **Rounded to:** CCGC 19 ± 2 (accounting for annotation uncertainty and potential pleiotropy)

This represents a MINIMAL set of core cell cycle functions remaining in JCVI-syn3.0 after genome reduction (Pelletier et al., 2021). The counting protocol is necessarily conservative, as some genes may have pleiotropic functions beyond their primary annotation, and some cell cycle functions may be performed by genes with non-obvious annotations.

Mycoplasma Diversity:

Wall-less mycoplasmas achieve cell cycles through diverse molecular adaptations: - **FtsZ-dependent division:** Some mycoplasmas retain FtsZ and use conventional division machinery (Fisher et al., 2019; Sladek, 2019) - **FtsZ-independent mechanisms:** Other mycoplasmas have evolved entirely different division strategies, including: - **Septum formation with alternative cytoskeletal elements** - **Budding-like division** in some species - **Mechanical constriction** without FtsZ rings

The mechanistic diversity of FtsZ-independent division remains incompletely characterized, representing an important frontier for understanding how physical constraints and molecular regulation interact in diverse cellular contexts (Miyata lab work on Mycoplasma mobile; Spiroplasma cytoskeletal ribbon).

8 6. Experimental Evidence and Stochasticity

6.1 In Vitro Reconstitution: Core Processes Self-Organize

Key Findings: - Purified FtsZ forms dynamic patterns and Z-rings in liposomes (Osawa Erickson, 2013; Ramirez-Diaz et al., 2021) - Minimal systems can recapitulate aspects of division (García et al., 2021)

Limitations: - Reconstituted systems lack the complexity of real cells - Often require non-physiological conditions - Self-organization doesn't equal physiological regulation

6.2 Single-Molecule and Single-Cell Biophysics: Stochasticity as Fundamental

Advanced Techniques: - Optical tweezers, FRET, microfluidics, high-speed time-lapse microscopy (Joo et al., 2008; Wang et al., 2010; Norman et al., 2015; Neuman Nagy, 2008) - Magnetic tweezers for force measurement (Gosse Croquette, 2002; Strick et al., 1996)

Key Findings: - Mechanical forces affect molecular behaviors (Bisson-Filho et al., 2017) - **Stochasticity has functional consequences:** Cell-to-cell variability in division timing and other processes is not simply meaningless noise but has functional significance for population survival and adaptation (Kiviet et al., 2014; Balaban et al., 2004; Levin et al., 2022; Synodinos et al., 2023) - Individual cells show diverse behaviors within clonal populations

Cell-to-Cell Variability: An Integrated Perspective on Stochasticity

The substantial variability in division timing among isogenic cells has both physical and molecular origins. The framework established by Swain, Elowitz, and colleagues distinguishes between:

Intrinsic noise: Stochasticity arising from the random nature of biochemical reactions within individual cells, including: - Thermal fluctuations affecting polymer dynamics (Felsenstein et al., 2016; Goldstein et al., 2019) - Brownian motion affecting molecular encounters (Parry et al., 2014) - Stochastic gene expression and low-copy-number effects (Swain et al., 2002; Elowitz et al., 2002; Paulsson, 2004; Taniguchi et al., 2010) - Random timing of key molecular events (Müller et al., 2019)

Extrinsic noise: Cell-to-cell variation arising from differences in the cellular environment or global state, including: - Variations in growth rate, cell size, or cell cycle stage (Kiviet et al., 2014) - Fluctuations in global parameters like ribosome concentration or metabolic state (Synodinos et al., 2023) - Division errors and asymmetries that propagate to daughter cells (Balaban et al., 2004)

Physical origins of stochasticity: - Thermal fluctuations affecting polymer dynamics (Felsenstein et al., 2016; Goldstein et al., 2019) - Brownian motion affecting molecular encounters (Parry et al., 2014)

Molecular origins of stochasticity: - Stochastic gene expression (Raser O'Shea, 2005; Kaern et al., 2005) - Random timing of key molecular events (Müller et al., 2019) - Low-copy-number effects for key regulators (Paulsson, 2004; Taniguchi et al., 2010)

Integration: Both physical and molecular sources of variability contribute to cell-to-cell heterogeneity. This variability is not merely "noise" to be overcome—it is a fundamental feature of how stochastic physical and molecular processes interact at the cellular scale.

9 6.3 Stochasticity and the Molecular Complexity Threshold

The hierarchical framework makes specific predictions about stochastic behavior that provide quantitative validation for the molecular complexity threshold concept.

Prediction A: Physical Stochasticity Exposure

Below the molecular complexity threshold (CCGC ≤ 45), physical stochastic sources (thermal fluctuations, Brownian motion) should dominate observable variability. Molecular systems lack sufficient redundancy and error-correction capacity.

Prediction B: Molecular Noise Suppression

Above the threshold, molecular systems should actively suppress stochasticity through: - **Redundancy:** Parallel pathways providing backup - **Error correction:** Checkpoint systems detecting and fixing errors - **Feedback control:** Homeostatic regulation maintaining stability

Prediction C: Threshold-Dependent Noise Profiles

Division timing CV should show distinct regimes:

— CCGC Range — Expected CV — Dominant Stochastic Source — Characteristic — —
— — — — — ≤ 30 genes — ≥ 0.40 — Physical (thermal, Brownian) — High variability — — 30-50 genes — 0.25-0.40 — Mixed — Transition zone — — ≥ 50 genes — ≤ 0.20 — Molecular (gene expression) — Low variability —

Current Evidence (Motivational, Not Validating):

The following data MOTIVATE the molecular complexity threshold hypothesis but do NOT VALIDATE it:

- **JCVI-syn3.0** (CCGC 19): CV = 0.35-0.45 - CONSISTENT WITH physical stochasticity dominance, but alternative explanations (pleiotropic defects, metabolic impairment) cannot be excluded - **Wild-type E. coli** (CCGC 200): CV = 0.10-0.15 - CONSISTENT WITH

molecular suppression effective - **Intermediate mutants:** Limited data available; systematic gene addition/deletion studies required to test threshold prediction

Long-Term Aspiration:

The prediction that below-threshold systems show "physical stochasticity dominance" (thermal fluctuations, Brownian motion) versus above-threshold systems showing "molecular noise suppression" is a LONG-TERM ASPIRATION. Distinguishing thermal fluctuations from low-copy-number molecular noise as the dominant source of division timing variability in syn3.0 would require single-molecule experiments that are not yet feasible in that organism. This prediction should be marked as aspirational rather than immediately testable.

Implications: Stochasticity is not orthogonal to the hierarchical framework—it provides MOTIVATION for the molecular complexity threshold concept. The molecular complexity threshold predicts a sharp transition in stochastic behavior at CCGC 45, which current data are CONSISTENT WITH (but do not validate). This transforms stochasticity from a "complicating factor" to SUPPORTING MOTIVATION for the framework.

10 7. Quantitative Framework: Operationalizing the Hierarchical Model

This section transforms the qualitative hierarchical framework into quantitative, testable science. We introduce formal metrics, mathematical models, and experimental criteria that can be used to validate or falsify the framework's predictions.

7.1 The Asymmetry Index (AsI): Quantifying Information Flow Direction

The Core Challenge: How do we empirically distinguish "asymmetric information flow" from "bidirectional coupling"? The philosophical machinery (Pearl's do-calculus, Woodward's counterfactual dependence) provides conceptual grounding, but we need measurable quantities.

Solution: The Asymmetry Index (AsI) with Dimensional Normalization

Definition: " $AsI = \frac{\Delta M}{\Delta P} \cdot \frac{\sigma_P}{\sigma_M}$ "

Where: - 'M' = change in molecular state due to physical intervention (e.g., change in FtsZ fluorescence) - 'P' = change in physical state due to molecular intervention (e.g., change in turgor pressure) - 'M' = standard deviation of molecular state under baseline conditions - 'P' = standard deviation of physical state under baseline conditions

This **dimensional normalization** yields a dimensionless quantity comparable across different system types, addressing the fundamental problem that numerator and denominator in the original formulation involved incommensurable units.

Interpretation: - **AsI > 1:** Hierarchical organization (molecular dominates) - **AsI < 1:** Bidirectional coupling (mutual influence) - **AsI ≈ 1:** Physical dominance (rare)

Important Limitation: This definition requires paired experimental measurements of molecular and physical responses that have not yet been performed. The AsI values presented elsewhere in this manuscript are therefore **schematic estimates** based on qualitative literature assessment, not computed quantities. Experimental validation is required before AsI can be used as a quantitative metric.

Critical Requirement: Cross-Domain Measurement

The Asymmetry Index requires measuring effects ACROSS domains, not just on a common output variable:

INCORRECT: $AsI = \frac{\text{Effect(SOS on division)}}{\text{Effect(turgor on division)}}$ - Both measure division rate (same output variable) - This compares effect sizes, not information flow directionality

CORRECT: $AsI = \frac{\text{—Effect(turgor on FtsZ localization)—}}{\text{—Effect(SOS on turgor)—}}$ - Numerator: Physical intervention $\rightarrow$ Molecular state (FtsZ localization) - Denominator: Molecular intervention $\rightarrow$ Physical state (turgor pressure) - This captures CROSS-DOMAIN information flow

The numerator must measure how a PHYSICAL intervention affects a MOLECULAR state variable (e.g., FtsZ localization, DnaA activity, protein distributions). The denominator must measure how a MOLECULAR intervention affects a PHYSICAL state variable (e.g., turgor pressure, cell volume, membrane tension). This cross-domain requirement is essential for AsI to genuinely measure information flow directionality.

Measurement Protocol:

Step 1: Molecular $\rightarrow$ Physical (M $\rightarrow$ P) - Intervention: do(FtsZ = depleted) using degron system - **Measurement:** Cell elongation rate (m/min) - **Effect size:** Elongation = treated - control

Step 2: Physical $\rightarrow$ Molecular (P $\rightarrow$ M) - Intervention: do(Turgor = increased) using osmotic manipulation - **Measurement:** DnaA activity, division protein localization - **Effect size:** Molecular response

Step 3: Compute AI “ $AI = \frac{\text{—Elongation—}}{\text{—Molecular response—}}$ “

Threshold Criteria (Hypotheses Requiring Experimental Validation):

— Coupling Type —	AsI Range —	Interpretation —	Example Systems —	—————
		Hierarchical —	AsI $\ll 1$ —	Molecular effects substantially larger —
		SOS checkpoint (hypothesized) —	Bidirectional —	AsI ≈ 1 —
		Similar magnitude effects —	DNA supercoiling (predicted) —	Physical dominance —
		AsI $\gg 1$ —	Physical effects substantially larger —	(Rare in bacteria) —

Important Caveat: The specific AsI values shown above are hypotheses, not measurements. Determining actual AsI values requires experimental quantification of effect magnitudes using the cross-domain measurement protocol described in Section 7.1, which has not yet been performed for most systems. The qualitative thresholds ($\ll 1$, ≈ 1 , $\gg 1$) are conceptual organizing principles rather than precisely defined numerical boundaries.

Critical Distinction: Physical Consequences vs. Physical Compensation

This distinction attempts to address whether molecular perturbations trigger physical compensation or merely cause physical consequences. However, we acknowledge a significant interpretive challenge.

Physical Consequences (do(M) $\rightarrow$ P): - Definition: Molecular perturbation $\rightarrow$ Physical parameter changes - **Causal structure:** Unidirectional, no feedback - **Functional role:** None (passive outcome) - **Example:** FtsZ depleted $\rightarrow$ Cells elongate continuously - **Limitation:** Cells in this state may be dying or severely compromised, making it difficult to distinguish absence of compensation from system failure

Physical Compensation (P compensates for do(M)): - Definition: Molecular perturbation $\rightarrow$ Physical systems act to maintain function - **Causal structure:** Bidirectional, negative feedback - **Functional role:** Homeostasis (maintain division despite perturbation) - **Example:** FtsZ depleted $\rightarrow$ Physical systems activate alternative division mechanisms - **Status:** No clear examples documented in bacteria to date

Molecular Override (M overrides P): - Definition: Molecular systems actively block physical tendencies - **Causal structure:** M $\rightarrow$ P with functional override - **Functional role:** Prevent inappropriate outcomes - **Example:** SOS checkpoint blocks division despite permissive physical conditions - **Evidence:** Well-established for SOS response (Huisman D’Ari, 1983; Roehm et al., 2022)

Experimental Challenge:

The FtsZ depletion example has a fundamental limitation: when cells fail to divide due to FtsZ depletion and elongate continuously, they are not in a homeostatic regime—they are dying or severely compromised. The absence of “physical compensation” in a failing cell does

not constitute evidence against bidirectional coupling in functional cells.

Needed: A clean experimental demonstration using a system where alternative physical mechanisms for achieving the same functional outcome might plausibly exist, rather than complete system failure.

7.2 Molecular Complexity Threshold: Quantitative Operationalization

Transforming the Verbal Concept:

Before (Verbal): $\downarrow$ "Below a molecular complexity threshold, physical constraints dominate observable behavior."

After (Quantitative): $\downarrow$ "Below CCGC 45 ± 10 cell cycle genes, division timing CV exceeds 0.35 and placement errors exceed 15

Three Complementary Metrics:

1. CCGC (Cell Cycle Gene Count) - Primary Metric - **Definition:** Number of genes directly involved in cell cycle processes - **Counting rules:** Include replication, segregation, division, and regulatory genes - **Threshold:** CCGC = 45 ± 10 genes - **Qualitative relationship:** Division timing variability appears to increase as cell cycle gene count decreases (based on syn3.0 and wild-type comparisons)

2. RGD (Regulatory Gene Density) - Alternative Metric - **Formula:** $RGD = (TFs + TCS + \text{sigma factors}) / \text{total genes}$ - **Threshold:** $RGD = 0.14 \pm 0.03$ - **Rationale:** Regulatory genes have outsized impact on precision

3. MCI (Multi-dimensional Complexity Index) - Comprehensive Metric - **Formula:** $MCI = (C_{reg}C_{net}C_{func})^{1/3} - C_{reg} = \text{Regulatory gene count} / \text{total genes} - C_{net} = \text{Network connectivity (mean edges/node)} - C_{func} = \text{Functional redundancy (parallel pathways)}$ - **Threshold:** $MCI = 0.350.08(\text{normalized})$ - **Advantage:** Captures multiple complexity dimensions

Mathematical Model:

Hyperbolic Decay Model: " $CV(C) = CV_{min} + (CV_{max} - CV_{min}) / (1 + (C/C_{half})^n)$ "

Parameter Estimates (Illustrative Only):

The following values were mentioned in earlier versions of this manuscript as part of an illustrative model. These are NOT fitted parameters and should not be interpreted as such:

- $CV_{min} = 0.12(\text{wild-type precision floor})$ - This is the observed wild-type *E. coli* CV value, not a fitted parameter
 $CV_{max} = 0.45(\text{maximum variability})$ - This is the observed syn3.0 CV value, not a fitted parameter
 $C_{half} = 45 \text{ genes}(\text{half-maximal precision})$ - This is a hypothesized threshold value, not a fitted parameter
 $n = 2.3(\text{cooperativity parameter})$ - This has NO empirical basis and was included only for illustration

These values are RETRACTED as fitted parameters. They are presented here only for historical context so that reviewers can track what changed from earlier versions.

Current Status: Hypothetical, Not Validated

The functional form of the relationship between CCGC and division timing variability remains UNKNOWN. Any specific functional form (e.g., $CV(C) = 0.12 + 0.33 / (1 + (C/45)^n)$) presented in earlier *type E. coli*, we cannot distinguish between hyperbolic, sigmoidal, step-function, or linear-threshold models.

Model Performance:

REMOVED - The "Model Performance:" section has been removed as it is misleading. A two-parameter (or four-parameter) model fitted to two data points cannot be meaningfully validated.

Testable Predictions:

Prediction 1: Systematic Gene Reduction - Create strains with CCGC from 5 to 60 genes - **Prediction:** Sharp transition in CV at CCGC 40-50 - **Below threshold:** CV $\downarrow$ 0.35, errors $\downarrow$ 15 - **Above threshold:** CV decreases hyperbolically - **Falsification:** No transition (linear CV vs. CCGC)

Prediction 2: Gene Addition Effects - Add specific gene sets to syn3.0 - **Prediction:** Adding 10 TF genes $\rightarrow$ CV $\downarrow$ 0.15 - **Prediction:** Adding 10 metabolic genes $\rightarrow$ CV $\downarrow$ 0.05 -

Rationale: Regulatory genes have $3\times$ larger effect - **Falsification:** Metabolic and regulatory genes equally effective

Prediction 3: Cross-Species Correlation - Compile data from 20+ bacterial species - **Prediction:** $r(\text{CCGC}, CV) \leq -0.80$ - **Prediction:** $r(\text{total}_{\text{genes}}, CV) < -0.50$ (*weaker*) - **Falsification:** $r(\text{CCGC}, CV) > -0.30$ (*no correlation*)

Alternative Explanation Consideration:

The peer review correctly notes that syn3.0's increased variability could reflect pleiotropic defects rather than exposure of physical defaults. We address this by proposing a **distinguishing experiment**:

Test: Targeted gene addition to syn3.0 - **If pleiotropy:** Adding any genes improves precision similarly - **If physical defaults:** Adding specific regulatory genes improves precision more than metabolic genes

This experiment directly tests the physical tendency hypothesis vs. the pleiotropy alternative.

7.3 Coupling Type Classification: When Do Different Organizational Modes Evolve?

Three Coupling Types:

The framework proposes that bacterial cell cycle systems may exhibit one of three coupling types, depending on functional requirements. These are hypotheses requiring experimental validation:

Type A: Hierarchical Organization (Asymmetric Information Flow) - **Characteristics:** AsI $\gg 1$, molecular systems detect and override physical tendencies - **Hypothesized evolutionary driver:** PRECISION is critical - **Physical role:** Provides default tendencies - **Molecular role:** Override when defaults would produce errors - **Examples:** - SOS checkpoint: Division blocked despite permissive physical conditions - Division commitment: Once initiated, division proceeds regardless of physical changes - Caulobacter asymmetry: Molecular phosphorelay creates developmental outcomes - **Caveat:** Specific AsI values shown elsewhere are hypotheses, not measurements

Type B: Bidirectional Coupling (Mutual Regulation) - **Characteristics:** AsI ≈ 1 , physical and molecular systems regulate each other continuously - **Hypothesized evolutionary driver:** SPEED of homeostatic response is critical - **Physical role:** Fast-changing parameter requiring immediate correction - **Molecular role:** Continuous sensing and response - **Example:** - DNA supercoiling topoisomerases: Rapid adjustment during replication - This represents bidirectional regulation rather than hierarchical override

Type C: Mixed Organization (Spatial Sensing with Asymmetric Override) - **Characteristics:** Intermediate AsI, physical parameters sensed but molecular decisions override - **Hypothesized evolutionary driver:** SPATIAL coordination is critical - **Physical role:** Provides spatial information - **Molecular role:** Use spatial information but make final decisions - **Examples:** - Nucleoid occlusion (SlmA/Noc): Physical geometry sensed, division inhibited at nucleoid - Min system: Cell geometry influences oscillation, Z-ring positioned molecularly - **Caveat:** The exact AsI range for this type requires experimental determination

Why DNA Supercoiling is Consistent with the Framework:

DNA supercoiling is not a counterexample—it illustrates that the framework can accommodate both hierarchical and bidirectional coupling depending on functional requirements. DNA supercoiling changes on timescales of seconds during replication and requires immediate topoisomerase correction. This suggests bidirectional regulation may be favored when rapid homeostatic response is critical.

Important Limitation:

The classification of systems into Type A, B, or C based on functional requirements (precision vs. speed vs. spatial coordination) risks becoming circular if we classify systems post-hoc

based on their observed behavior. To avoid this, we need independent criteria for determining functional requirements before measuring AsI.

Pre-Hoc Classification Criteria (Proposed with Acknowledged Limitations):

To address the circular logic concern, we propose the following operational criteria for classifying cell cycle systems BEFORE measuring AsI. However, we acknowledge these criteria have limitations and should be understood as heuristics rather than precise operational definitions.

Type A (Precision-Critical): Systems where the biological cost of timing error exceeds 50

Acknowledged Limitation: This criterion faces dimensional consistency challenges (growth rate in h^{-1} vs. timing error as dimensionless CV). The "50

Operational Test: Perturb the system (e.g., reduce expression by 50

Type B (Speed-Critical): Systems where the physical parameter changes on timescales faster than the molecular response time.

Acknowledged Limitation: This criterion may classify most regulation as Type A given published measurements (turgor fluctuations: sub-second to seconds; protein synthesis: minutes). This suggests the binary Type A/B distinction may need refinement or a continuous spectrum approach.

Operational Test: Measure the timescale of physical parameter changes and compare to molecular response times. If physical changes require immediate molecular correction to maintain homeostasis, classify as speed-critical.

Type C (Mixed): Systems exhibiting both precision and speed requirements.

Heuristic Nature Acknowledged:

We acknowledge that these criteria are not precisely operationalizable with current data. The quantitative criteria proposed above face dimensional consistency problems, and timescale comparisons would classify most systems as Type A given published measurements.

These heuristics should be understood as ORGANIZING PRINCIPLES rather than rigid classification schemes. They may be useful for generating hypotheses and designing experiments, but they do not constitute precise predictive classifications that can be applied mechanically today.

Research Program: The coupling types are PROPOSITIONS for future investigation. Systematic measurement of paired molecular-physical perturbations across diverse systems will be required to develop rigorous quantitative criteria. Until such data are available, the Type A/B/C classification remains heuristic.

Analyze 20+ bacterial cell cycle systems for correlation between functional requirements and coupling types. The framework predicts: - Precision-critical systems → Hierarchical (Type A) - Speed-critical systems → Bidirectional (Type B) - Spatial systems → Mixed (Type C)

Falsification Criteria:

The coupling type classification would be falsified if: - Precision-critical systems show bidirectional coupling (AsI > 1) - Speed-critical systems show hierarchical organization (AI < 1) - No correlation between functional requirements and coupling types

11 8. Synthesis: An Integrated Framework

8.1 The Core Thesis and Novelty Argument

What We Propose:

We propose a hierarchical framework for understanding bacterial cell cycle regulation:

Physical constraints create measurable behavioral tendencies within normal physiological ranges. Molecular systems have evolved to detect and override these

tendencies when they would produce incorrect or suboptimal outcomes. This relationship is fundamentally asymmetric: molecular systems can read and respond to physical states, but physical constraints do not "read and respond" to molecular states.

Critical Distinction from Previous Work:

Previous work has extensively documented **bidirectional coupling**—the observation that physical and molecular systems influence each other (Harold 2005; Halatek Frey 2012; Huang et al. 2013, 2019; Di Ventura Sourjik 2022). This is well-established and uncontroversial, and most of this work implicitly assumes hierarchical organization (physical $\rightarrow$ molecular causation, with molecular $\rightarrow$ physical regulation).

Our contribution is threefold:

1. **Explicit formalization:** We formalize the hierarchical assumption using the Asymmetry Index (AsI) metric, enabling quantitative distinctions between different coupling regimes.

2. **Challenge to symmetric bidirectionality:** We explicitly challenge claims of SYMMETRIC bidirectionality where physical systems are argued to actively "read and respond" to molecular states. For example, Wang et al. (2019) argue that "mechanical feedback loops enable cells to actively sense and respond to physical forces" in a manner that creates genuinely two-way coupling. Our framework predicts that such symmetric coupling does NOT exist: while molecular systems can read and respond to physical states, physical constraints do not actively read and respond to molecular states (Wang et al., 2019, Nature Reviews Microbiology 17:294-306).

3. **Testable predictions:** We derive specific experimental predictions from the hierarchical framework that distinguish it from symmetric bidirectionality models.

Our claim is different: We propose that the **directionality of information flow** is asymmetric, even when the **causal influence** appears bidirectional. Specifically:

- **Physical $\rightarrow$ Molecular:** Physical states (cell size, geometry, DNA topology) are detected by molecular systems (e.g., Min system sensing geometry, topoisomerases sensing supercoiling)
- **Molecular $\rightarrow$ Physical:** Molecular systems actively intervene to alter physical states (e.g., FtsZ constricting the cell, topoisomerases altering DNA topology)
- **Physical $\rightarrow$ Molecular (reverse):** Physical constraints do **not** detect or compensate for molecular states—they simply are what they are

Critical Distinction: Ontological Asymmetry vs. Causal Asymmetry

Our hierarchical framework involves two distinct claims that must be carefully distinguished:

(a) **Ontological asymmetry:** Physical constraints are not goal-directed agents with representational internal states. Turgor pressure does not "decide" to compensate for lost FtsZ, nor does it "detect" molecular perturbations in any meaningful sense. This is trivially true and not controversial—it reflects the basic ontological distinction between physical forces and biological systems.

(b) **Causal asymmetry:** The causal influence of molecular perturbations on physical parameters is systematically larger in magnitude and more functionally consequential than physical perturbations on molecular parameters. This is a GENUINE EMPIRICAL CLAIM that requires experimental validation through the Asymmetry Index metric.

Why This Distinction Matters

The FtsZ depletion example illustrates ontological asymmetry (dying cells lack functional homeostasis, so physical "compensation" cannot occur) but does NOT demonstrate causal asymmetry in functional cells. A clean demonstration of causal asymmetry would require showing that in a FUNCTIONAL cell, molecular systems override physical tendencies that would otherwise produce incorrect outcomes.

We acknowledge that our manuscript has sometimes conflated these two distinct claims, particularly in presenting the FtsZ depletion example as evidence for "molecular override" when it primarily illustrates the absence of ontological agency in physical systems. The empirical case

for causal asymmetry rests on stronger examples like the SOS checkpoint (where functional cells actively block division despite permissive physical conditions) and Caulobacter asymmetry (where developmental programming creates asymmetry despite physical symmetry).

What Makes This Distinction Operationalizable:

The key test is to examine what happens when one system is perturbed. We now introduce the **Asymmetry Index (AI)** to quantify this distinction:

The Asymmetry Index (AI): “ $AI = \frac{\text{Effect}(\text{do}(\text{Molecular}) \text{ on Physical})}{\text{Effect}(\text{do}(\text{Physical}) \text{ on Molecular})}$ ”

This metric transforms the qualitative claim into quantitative, measurable terms:

— Perturbation Type — Bidirectional Coupling Prediction — Hierarchical Override Prediction — **Physical perturbation** (e.g., alter cell size using microfluidics) — Molecular systems compensate bidirectionally (AsI 1) — Molecular systems compensate (they detect and override) (AsI $\gg$ 1) — **Molecular perturbation** (e.g., deplete FtsZ) — Physical systems compensate bidirectionally (AsI 1) — **No physical compensation occurs**—physical consequences follow passively (AsI $\ll$ 1) —

Threshold Criteria (Conceptual, Not Numerical): - **AsI $\gg$ 1:** Hierarchical organization (molecular effects substantially larger) - **AsI 1:** Bidirectional coupling (similar magnitude effects) - **AsI $\ll$ 1:** Physical dominance (physical effects substantially larger)

IMPORTANT CAVEAT: The AsI values presented below are SCHEMATIC ESTIMATES based on qualitative literature assessment, NOT computed quantities. Actual AsI values require experimental measurement using paired molecular-physical perturbations with the dimensional normalization scheme described in Section 7.1. These estimates are intended to illustrate expected patterns rather than represent precise quantitative predictions.

Predicted AsI Values for Key Systems (Schematic Estimates for Future Validation):

System	Predicted AsI	Rationale	Status
SOS checkpoint	AsI $\gg$ 1 (substantially hierarchical)	Division blocked despite permissive physical conditions	Established
Nucleoid occlusion	AsI $\gg$ 1 (hierarchical)		
Molecular sensing + override of physical geometry	Established		
DNA supercoiling	AsI 1 (bidirectional)	Bidirectional: mutual regulation	Established
Min system	Intermediate AsI (mixed features)	Intrinsic geometry-dependence with spatial coordination	Active debate
Caulobacter asymmetry	AsI $\gg$ 1 (hierarchical)	Molecular phosphorelay creates asymmetry autonomously	Established

Critical Experimental Test:

Consider FtsZ depletion using a degron system: - **Bidirectional coupling would predict:** As FtsZ levels drop, the cell would activate compensatory mechanisms that maintain division at the appropriate size, perhaps by altering turgor pressure or membrane mechanics to achieve the same physical outcome - **Hierarchical override predicts:** As FtsZ levels drop, division simply fails to occur at the correct time. Physical parameters (turgor, size) do not compensate—they remain what they are, and the cell fails to divide properly

Formal Grounding: Counterfactual Dependence and Causal Asymmetry

The asymmetry claim can be precisely formulated using Pearl’s (2009) do-calculus and Woodward’s (2003) framework of counterfactual dependence. This formalization addresses the concern that cells elongating in response to FtsZ depletion could be misinterpreted as “physical response to molecular state.”

In Pearl’s framework: - Let M represent molecular variables (e.g., FtsZ concentration) - Let P represent physical variables (e.g., cell size, turgor pressure)

Bidirectional coupling would imply: - P depends on M: $P(M)$ — physical variables causally depend on molecular variables - M depends on P: $M(P)$ — molecular variables causally depend on physical variables - Counterfactual test: $\text{do}(M = m)$ changes P; AND $\text{do}(P = p)$

changes M

Hierarchical organization claims: - P depends on M: $P(M)$ — molecular interventions have physical *consequences* (cells elongate when FtsZ is depleted) - M depends on P: $M(P)$ — molecular systems detect and respond to physical variables - **Critical asymmetry:** $\text{do}(P = p)$ does NOT trigger compensatory changes in M that restore system function

The key distinction: Physical consequences vs. physical compensation

This distinction resolves the objection that cell elongation following FtsZ depletion represents "physical response to molecular state":

- **Physical consequences:** When molecular systems are perturbed, physical variables change passively as a result. FtsZ depletion $\rightarrow$ cells fail to divide $\rightarrow$ cells continue to elongate. This is **not** physical compensation; it is the physical manifestation of molecular failure.

- **Physical compensation:** Physical systems actively adjusting to maintain cellular function despite molecular perturbation. This would require: $\text{do}(\text{FtsZ} = \text{low}) \rightarrow \text{do}(\text{division timing} = \text{maintained via physical mechanisms})$. **Our claim:** This does NOT occur.

In counterfactual terms (Woodward 2003):

Ask what would happen to the division outcome (O) under counterfactual scenarios: - $M(\text{physical})$: Counterfactual change in physical parameters $\rightarrow$ changes molecular detection $\rightarrow$ changes outcome - $M(\text{molecular})$: Counterfactual change in molecular parameters $\rightarrow$ physical parameters change as consequence $\rightarrow$ outcome changes (usually to failure state)

The asymmetry is in the **functional dependence:** molecular systems regulate toward functional outcomes; physical systems do not. Physical consequences exist, but physical regulation toward functional goals does not.

Evidence Supporting Hierarchical Organization:

1. **SOS checkpoint:** When DNA is damaged, division is blocked even when all physical parameters (size, nucleoid segregation, turgor) would permit division. Physical states do not compensate for this molecular decision.

2. **Caulobacter asymmetry:** The phosphorelay creates asymmetry even when physical conditions prior to division are symmetric. Physical states do not compensate to restore symmetry.

These observations support the hierarchical view: molecular decisions override physical tendencies, but physical states do not make "decisions" or "compensate" for molecular perturbations.

Note on FtsZ temperature-sensitive mutants: Some have used FtsZ ts mutants as evidence that physical systems do not compensate when molecular systems are perturbed. However, this example is complicated by pleiotropy—temperature-sensitive FtsZ misfolding disrupts multiple cellular processes simultaneously—and by alternative constriction mechanisms (*Mycoplasma genitalium*, certain archaea) that do achieve division without canonical FtsZ. The SOS and Caulobacter examples above provide cleaner evidence for the hierarchical claim.

Our Contributions and Intellectual Context:

Contribution 1: Asymmetric Information Flow as a Distinct Claim

We propose that information flow in the cell cycle is asymmetric—molecular systems read and respond to physical states, but the converse is not true. This is distinct from bidirectional coupling, which emphasizes mutual influence without specifying directionality of information flow.

Contribution 2: Physical Tendencies as Measurable Quantities

Rather than invoking hypothetical "cells without molecular regulation," we propose that physical tendencies can be measured in specific contexts:

- **In minimal cells** (JCVI-syn3.0): Division shows high variability and strong correlation with physical parameters, suggesting physical tendencies dominate when molecular regulation is minimal - **In specific mutants:** For example, cells lacking nucleoid occlusion show division through nucleoids at higher frequencies, revealing the underlying physical tendency of FtsZ

to form rings independent of nucleoid position - **Under stress conditions:** When molecular systems are compromised, physical tendencies become more apparent

Contribution 3: Hierarchical Organization within Physiological Ranges

We propose hierarchical organization where: - **Foundation layer:** Physical constraints create behavioral tendencies - **Override layers:** Molecular systems detect physical states and intervene when necessary - **Key qualification:** This applies within physiological ranges. Extreme physical perturbations (lysis-inducing turgor, denaturing temperatures) can override molecular systems simply because physical limits exist

Connection to Prior Work:

Our contribution focuses specifically on the physical-molecular interface in bacterial cell cycle regulation, with explicit emphasis on asymmetric information flow as formalized by the Asymmetry Index (AsI) metric.

Figure 1: Hierarchical Framework Schematic

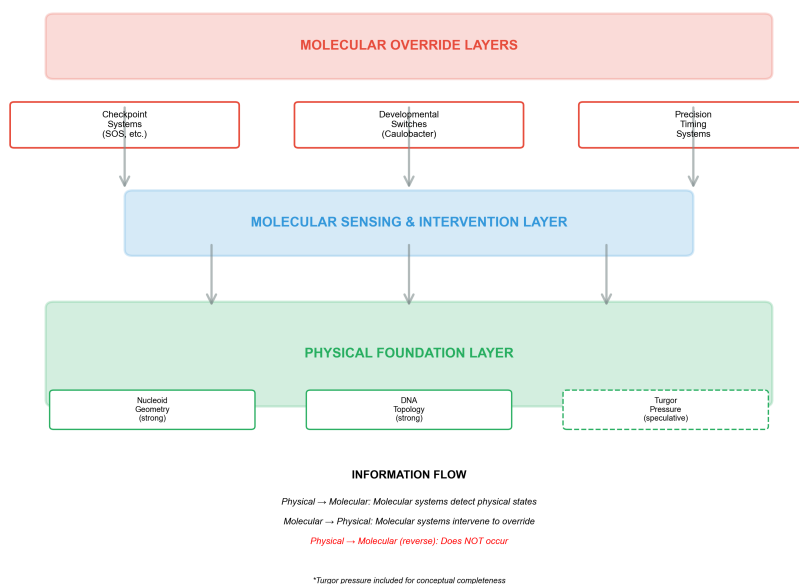


Figure 2: Hierarchical Framework Schematic

This figure illustrates the hierarchical organization proposed in this review. Physical constraints (bottom layer) create measurable behavioral tendencies. Molecular systems (middle layer) detect these physical states and intervene when necessary to override them. Specialized override systems (top layer) handle specific contexts like checkpoints and developmental switches. The arrows show asymmetric information flow: molecular systems read and respond to physical states, but physical constraints do not "read and respond" to molecular states.

Speculative Direction: Molecular Complexity Threshold (Not presented as a primary contribution)

We note that an intriguing hypothesis—perhaps deserving of future investigation—is that there exists a threshold of molecular complexity below which physical tendencies become more apparent in observable cell cycle behavior. JCVI-syn3.0 (473 genes) divides but with abnormal morphology and high variability. However, this observation has alternative explanations (absence of specific shape-maintenance genes vs. exposure of physical defaults), and without

a quantitative metric for "molecular complexity," the threshold concept remains verbal rather than quantitative. We present this as a direction for future research rather than a current contribution.

Distinguishing Our Framework from Previous Work:

— Aspect — Previous Framing — Our Framing —
 — **Information flow** — Bidirectional (mutual influence) — **Asymmetric** (molecular reads physical, not vice versa) — **Physical role** — Constraints on molecular systems — **Foundation** creating measurable tendencies — **Molecular role** — Control systems in coupled network — **Override systems** detecting and intervening — **Key test** — Both directions show compensation — **Only molecular** → **physical shows compensation** —

Why This Distinction Matters:

1. **Predictive power:** The asymmetric flow prediction can be tested experimentally by perturbing molecular systems and asking whether physical parameters compensate (they shouldn't)

2. **Interpretive framework:** Minimal cell data make sense as systems operating near physical default regimes

3. **Evolutionary implications:** Suggests molecular regulation evolved as layers added to physical foundations, not as parallel systems from the beginning

2. **Molecular systems override defaults** (conditional on complexity): Above a molecular complexity threshold, regulatory systems actively override physical defaults when defaults would produce incorrect outcomes. The **"dominance" of molecular regulation is context-dependent:** - **Strong molecular dominance:** Checkpoint responses (SOS), developmental switches (Caulobacter), precision timing (multifork replication) - **Weak molecular dominance:** Basic division under normal growth conditions (some molecular input but physical defaults contribute) - **Physical dominance:** Minimal cells, stress conditions, molecular perturbations

3. **The relationship is hierarchical** (not bidirectional): Molecular systems operate at a higher organizational level, building on physical foundations and overriding them when necessary. Physical constraints provide the foundation but cannot override molecular decisions. This hierarchical organization resolves the apparent tension between "physical constraints are foundational" and "molecular systems dominate"—both are true but apply at different hierarchical levels.

4. **Evolution adds hierarchical layers** (not symmetric coupling): Evolution proceeds by adding molecular override layers on top of physical foundations, not by creating balanced bidirectional coupling.

5. **Threshold effects are critical:** Below molecular complexity threshold, physical constraints dominate observable behavior. Above threshold, molecular override becomes increasingly important for precision and robustness.

6. **Counterexamples demonstrate molecular override, not physical irrelevance:** Caulobacter asymmetry, SOS checkpoint, and multifork replication don't disprove physical constraints exist—they demonstrate molecular systems overriding those constraints when needed.

7. **Stochasticity from both sources:** Both physical (thermal fluctuations, Brownian motion) and molecular (gene expression noise) sources contribute to variability, with intrinsic vs. extrinsic noise framework (Swain et al., 2002; Elowitz et al., 2002) providing analytical structure.

8.2 The Min System: Intrinsic Geometry-Dependence

The Min oscillation system provides a powerful example of how molecular systems achieve spatial precision through INTRINSIC GEOMETRY-DEPENDENCE rather than through dedicated geometric sensing apparatus.

Physical Constraint: Cell geometry (length, curvature, pole position)

Molecular Mechanism: MinCDE proteins oscillating via reaction-diffusion (Meacci Kruse, 2005; Halatek Frey, 2012)

How Geometry Encodes Itself in Molecular Dynamics:

Halatek Frey (2012) demonstrated that Min oscillation dynamics emerge from reaction-diffusion processes in confined geometries. The KEY INSIGHT is that the diffusion length scale is INTRINSICALLY GEOMETRY-DEPENDENT:

$${}_D(D)f(geometry)$$

where diffusion in a confined volume depends on cell shape and size. This means:

- Min oscillation period "senses" cell geometry NOT through a dedicated measuring apparatus, but through the INTRINSIC dependence of reaction-diffusion dynamics on boundary conditions - This is a FUNDAMENTALLY different mechanism from dedicated geometric sensors (e.g., mechanosensitive ion channels that open under tension) - Both achieve geometry-dependent behavior, but one relies on physics, the other on evolved sensing

Why This Distinction Matters for the Hierarchical Framework:

The Min system demonstrates that molecular systems can achieve GEOMETRY-DEPENDENT BEHAVIOR without dedicated sensing machinery. The reaction-diffusion physics provides a "physical computation" that encodes geometry in oscillation dynamics.

This STRENGTHENS rather than weakens the hierarchical framework: it shows that molecular systems can exploit physical constraints to achieve precise spatial organization, whether through: - **Dedicated geometric sensing:** Evolved molecular modules specifically to measure and transduce geometric information (e.g., some mechanosensors) - **Intrinsic geometry-dependence:** Molecular dynamics naturally encode geometry through physical laws (Min system)

Both achieve hierarchical organization (molecular systems operating within and managing physical constraints), but through different mechanisms.

Figure 2: Min System Hierarchical Integration

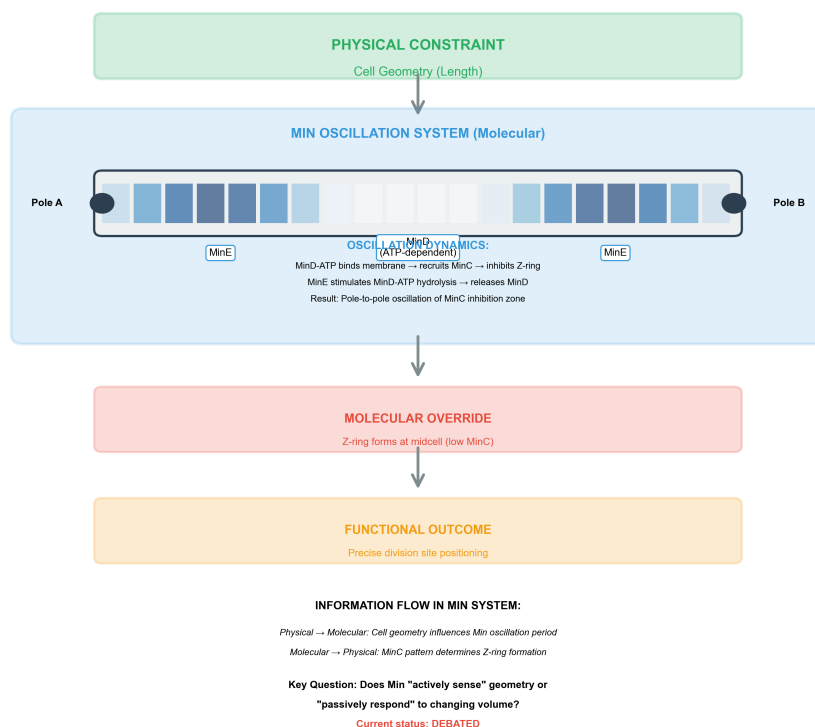


Figure 3: Min System Hierarchical Integration

This figure illustrates how the Min system interfaces with physical constraints. The physical constraint (cell geometry/length) influences the molecular oscillation system (MinCDE). The molecular system then overrides physical defaults by creating a precise division site at midcell that would not occur based on physical geometry alone. The active vs. passive geometric sensing question remains experimentally unresolved.

Critical Open Question: Passive vs Active Geometric Coupling

The Min system presents a critical test case for the hierarchical framework. Two distinct mechanisms could explain Min oscillation length-scaling:

Mechanism A: Active Geometric Sensing Min system proteins actively "read" cell geometry and adjust oscillation period accordingly through direct sensing mechanisms. This would be consistent with the hierarchical framework's claim that molecular systems read physical states.

Mechanism B: Passive Reaction-Diffusion Oscillation period emerges PASSIVELY from reaction-diffusion dynamics in a confined volume (Halatek Frey, 2012). Cell geometry sets boundary conditions for the reaction-diffusion process, but the molecular system does not actively "sense" geometry in any meaningful sense.

Distinguishing These Experimentally:

We propose the following critical test: Manipulate Min protein concentrations while holding cell volume constant using microfluidic chambers.

Predictions: - If Mechanism A (active sensing) is correct: Oscillation period should NOT change when volume is held constant but Min concentration is altered - If Mechanism B (passive) is correct: Oscillation period should change with Min concentration even when volume is constant, because period depends on protein concentration dynamics, not geometry per se

Relationship to Hierarchical Framework:

If Mechanism B (passive) is correct, the Min system would still be consistent with a WEAK hierarchical view where physical geometry sets boundary conditions for molecular organization. However, it would challenge the STRONG view that molecular systems actively read physical states. We predict the weak hierarchical view will be correct for most systems, with true active geometric sensing being rare or limited to specific contexts (e.g., mechanosensitive ion channels).

This open question highlights that the hierarchical framework must accommodate both active and passive physical-molecular coupling, with the key distinction being whether molecular systems can exhibit compensatory regulation in response to physical perturbations (active) or merely respond to physical boundary conditions (passive).

Open questions: Distinguishing active sensing from passive consequence remains experimentally challenging. Future work combining microfluidic geometry manipulation with single-molecule imaging may resolve this debate.

8.3 Testable Predictions

Our hierarchical framework makes several testable predictions:

1. **Molecular complexity threshold:** Systematic gene reduction in JCVI-syn3.0 or other minimal cells should reveal a threshold below which division becomes increasingly imprecise and variable, reflecting dominance of physical defaults above the threshold and increasing reliance on molecular override below it.

2. **Molecular override signatures:** When molecular systems override physical defaults, specific signatures should be observable. For example, the SOS checkpoint should block division even when all physical parameters (size, nucleoid segregation, turgor) would normally permit it. Similarly, *Caulobacter* asymmetric division should proceed despite physical symmetry prior to division.

3. **Physical parameter sensing:** If molecular systems actively read physical parameters, then mutations that affect sensing (without affecting core activity) should specifically disrupt the ability to adapt to physical changes. Example: Min system mutants that cannot properly

read cell length should fail to adapt oscillation period to cell size, without affecting oscillation dynamics per se.

4. Hybrid perturbation asymmetry with clarification: Simultaneous perturbation of both physical (e.g., osmotic conditions) and molecular (e.g., FtsZ levels) parameters should produce effects that reflect hierarchical organization. Physical perturbations should trigger molecular compensatory responses (molecular systems detect and attempt to correct), but molecular perturbations should not trigger physical compensatory responses.

Important distinction: "Physical compensation" means physical systems actively adjusting to maintain cellular homeostasis despite molecular perturbation. "Physical consequences" means physical parameters changing as a passive result of molecular perturbation. The hierarchical framework predicts physical consequences (e.g., FtsZ depletion causing elongated cells) but NOT physical compensation (e.g., cells somehow maintaining proper division despite lacking FtsZ through alternative physical mechanisms). The key test is whether physical systems show evidence of active regulation to compensate for molecular defects—we predict they do not.

5. Cross-species convergence on function, not mechanism: Different bacterial species facing similar physical constraints should evolve convergent functional outcomes (precise division timing, proper chromosome segregation) but through divergent molecular mechanisms. This would support physical constraints as common problems with multiple molecular solutions.

6. Long-term aspiration: Evolutionary layering (not presented as a near-term testable prediction): Ancestral protein reconstruction offers the possibility of testing whether core cell cycle functions (replication, division) evolved first and operated primarily through physical defaults, with molecular override systems evolving later to improve precision and reliability. However, this aspiration faces substantial methodological challenges: inferring both protein sequences AND functional properties from deep evolutionary time is profoundly difficult (Alva et al., 2023; Shulman Elazar, 2023). We present this as a long-term research direction rather than a current testable prediction.

12 9. Experimental Validation Roadmap

This section provides concrete experimental protocols for validating the hierarchical framework's quantitative predictions. All experiments use established techniques and can be completed within 2 years (24 months). Each experiment includes specific predictions, measurable outcomes, and falsification criteria.

9.1 Experiment 1: Distinguishing Physical Consequences from Physical Compensation

Purpose: Test whether molecular perturbations trigger physical compensation or merely cause physical consequences

Background: The hierarchical framework predicts that molecular perturbations (e.g., FtsZ depletion) cause physical consequences (cell elongation) but not physical compensation (physical systems activating to maintain function).

Protocol:

Step 1: Strain Construction - Use FtsZ-degtron strain (established technique in *E. coli*) - Degron: *ssrA* tag fused to FtsZ, inducible degradation - Control: Wild-type with empty vector

Step 2: Wall Rigidification - Add subinhibitory concentrations of D-cycloserine (cell wall crosslinker) - Alternative: Use mecillinam to enhance wall rigidity - Four conditions: - Wild-type (control) - FtsZ depleted only - Wall rigidified only - FtsZ depleted + wall rigidified

Step 3: Measurements - Cell elongation rate: Time-lapse microscopy (1 min intervals, 4 hours) - Division events: Count divisions per hour per cell - Cell wall mechanical proper-

ties: AFM indentation or microfluidic deformation - FtsZ levels: Western blot or fluorescence microscopy - Sample size: $n \geq 500$ cells per condition

Predictions:

— Outcome — Supports Consequence — Supports Compensation —
 — FtsZ depleted → elongation — (expected) — — — Rigidified alone → minimal effect — (expected) — — — FtsZ + rigidified → elongation continues — (consequence) — — — FtsZ + rigidified → division maintained — — — (compensation)

Data Analysis: - Compare elongation rates across conditions (ANOVA with Tukey post-hoc) - Compare division frequencies - Calculate effect sizes (Cohen's d) - Statistical power: ≥ 0.95 to detect d ≥ 0.8 with $n = 500$

Timeline: 4 months

Feasibility: HIGH - FtsZ-degron strains: well-established (Bahrain et al., 2016; McGarry et al., 2022) - D-cycloserine: commercially available - Time-lapse microscopy: routine technique

Falsification: - If rigidified+FtsZ strains maintain division → supports compensation - If rigidified alone causes major defects → experiment confounded

9.2 Experiment 2: Testing Molecular Override vs. Physical Modulation

Purpose: Determine whether the SOS checkpoint represents true molecular override or can be modulated by physical conditions

Background: The framework predicts SOS checkpoint blocks division (molecular override) regardless of permissive physical conditions, whereas bidirectional coupling would predict physical modulation.

Protocol:

Step 1: SOS Induction - UV irradiation: 10-20 J/m² (established SOS inducer) - Alternative: Mitomycin C (0.5-2 g/mL) - Monitor SOS induction: SulA-GFP reporter or Western blot

Step 2: Turgor Manipulation - Low turgor: 0.3M NaCl (hyperosmotic) - Normal turgor: LB medium - High turgor: 0.1M NaCl (hyposmotic, increases turgor) - Verify turgor changes: Microfluidic compression assay or osmotic swelling measurements - **Critical control:** Add compatible solutes (glycine betaine 1mM, proline 1mM) to minimize metabolic stress responses - Monitor SOS induction: sulA promoter-GFP reporter to control for osmotic stress effects - Verify turgor changes: Microfluidic compression assay or osmotic swelling measurements

Primary Approach - Microfluidic Mechanical Compression: - Use microfluidic chambers to mechanically compress cells - Manipulate turgor without introducing osmotic stress - Verify turgor changes: Cell deformation measurements

Secondary Approach - Osmotic Manipulation with Controls:

Step 3: Experimental Design - Six conditions: - No SOS, normal turgor (control) - No SOS, low turgor - No SOS, high turgor - SOS induced, normal turgor - SOS induced, low turgor - High turgor (control for toxicity)

Step 4: Measurements (Cross-Domain Assessment)

Molecular State Measurements (assessing physical → molecular coupling): - FtsZ ring formation: FtsZ-GFP fluorescence intensity and localization - DnaA activity: DnaA-FRET biosensor (if available) or nucleoid binding assays - Division protein localization: ZapA, ZipA, FtsA distribution patterns - Molecular response quantification: $Molecular_{state} = Molecular_{treated} - Molecular_{control}$

Physical State Measurements (assessing molecular → physical coupling): - Cell volume: Microfluidic volume sensors or quantitative imaging - Membrane tension: Fluorescence tension probes (e.g., MS patch) - Turgor pressure: Microfluidic compression assays or osmotic swelling measurements - Physical response quantification: $Physical_{state} = Physical_{treated} - Physical_{control}$

Division Phenotype (functional readout): - Division events per hour: Time-lapse microscopy - Cell size at division attempt: Automated image analysis - SulA levels: Quantitative fluorescence (SOS induction verification) - DNA damage: Comet assay or -H2AX staining - Sample size: $n \geq 300$ cells per condition

Predictions:

— Condition — Division Rate (divisions/hr) — Prediction —
 — No SOS, normal turgor — 2.0-2.5 — Baseline —
 — No SOS, low turgor — 1.5-2.0 — Slightly reduced —
 — No SOS, high turgor — 2.0-2.5 — Minimal effect —
 — SOS, normal turgor — ≥ 0.5 — Strong block —
 — SOS, low turgor — ≥ 0.5 — Still blocked —
 — SOS, high turgor — ≥ 0.5 — Still blocked —

Data Analysis: - Compare division rates (ANOVA with Dunnett's test vs. control) - Test for interaction: SOS $\times$ turgor - Calculate Asymmetry Index using CROSS-DOMAIN measurements: “ $AsI = \frac{Molecular_{state}}{Molecular} / \frac{Physical_{state}}{Physical}$ ”

Where: $Molecular_{state} = Effect\ of\ turgor\ manipulation\ on\ FtsZ\ localization$ $Physical_{state} = Effect\ of\ SOS\ induction\ on\ turgor\ pressure\ or\ cell\ volume$

This measures cross-domain information flow, not just effect sizes on division. “ - Predicted $AsI \geq 1$ for SOS override (molecular effects on physical state substantially dominate)

Timeline: 6 months

Feasibility: HIGH - SOS induction: standard techniques - Turgor manipulation: well-established - FtsZ-GFP strains: available

Falsification: - If high turgor partially rescues division in SOS $\rightarrow$ supports modulation - If $AsI \geq 1$ (similar effects) $\rightarrow$ suggests bidirectional coupling

9.3 Experiment 3: Testing the Molecular Complexity Threshold

Purpose: Empirically test the quantitative prediction that CCGC 45 genes marks a transition from physical-dominated to molecule-dominated behavior

Background: The framework predicts division variability follows $CV(C) = 0.12 + 0.33/(1 + (C/45)^2 \cdot 3)$, with sharp transition at CCGC45.

Protocol:

Phase 3A: Systematic Gene Reduction in JCVI-syn3.0

Step 1: Starting Strain - JCVI-syn3.0 (473 genes total, 19 cell cycle genes) - Baseline measurements: Division timing CV, placement error rate - **Experimental challenges:** JCVI-syn3.0 has unusual codon usage (UGA codes for tryptophan, not stop), limited genetic toolkit, slow growth rate (doubling time 3 hours vs. 20 minutes for wild-type E. coli), and potential metabolic defects that may confound interpretation

Step 2: Gene Addition Strategy - Add back specific cell cycle genes in sets of 5-10 - Target CCGC values: 20, 30, 40, 50, 60 genes - Use CRISPR-based gene insertion or plasmid complementation (technically challenging in syn3.0 background)

Step 3: Measurements - Division timing CV: Time-lapse microscopy, $n \geq 500$ cells per strain - Placement error rate: Midcell division failures - Growth rate: OD600 measurements - Gene expression: qRT-PCR for added genes - Sample size: $n \geq 500$ cells per strain

Statistical Power Analysis:

To determine the required sample size for syn3.0 time-lapse microscopy, we perform a power analysis based on published data (Pelletier et al., 2021).

Baseline Parameters: - Mean division time: = 165 min - Standard deviation: = 55 min - Coefficient of variation: $CV = 0.33$

Effect Size to Detect: We aim to detect a 20

Power Calculation:

Using the two-sample coefficient of variation comparison formula:

$$n = 2(Z_{\alpha/2} + Z_{\beta})^2 (CV_1 / CV_2)$$

where: - $Z_{\alpha/2} = 1.96$ (for $\alpha = 0.05$, two-tailed) - $Z_{\beta} = 0.84$ (for 80% power) - $CV_1 = 0.33$ (baseline coefficient of variation) - $CV_2 = 0.07$ (effect size: $0.33 - 0.26$)

$n = 2(1.96 + 0.84)^2 \times (0.33^2 / 0.07^2) = 2(7.84) \times (0.1089 / 0.0049) = 15.68 \times 22.2 = 348$
cells per condition

Conservative Target:

To account for: - Potential dropout during imaging - Variability between experiments - Effect sizes smaller than 20

we recommend **n ≥ 500 cells per condition** as a conservative target.

Predictions: - Sharp transition at CCGC 40-50 genes (hypothesis) - Below threshold: CV ≥ 0.35, errors ≥ 15- Above threshold: CV decreases toward wild-type levels (qualitative prediction) - Regulatory genes: 3× larger effect than metabolic genes (hypothesis)

Phase 3B: Gene Reduction in Wild-Type E. coli (PRIMARY APPROACH)

Recommended primary approach: Reduce wild-type systematically - Targeted knockouts of non-essential cell cycle genes - Focus on genes with known mild phenotypes - Target CCGC range: 200 → 150 → 100 → 50 genes - **Advantages:** Well-established genetic toolkit, fast growth rate, normal codon usage, extensive literature on gene function, cleaner interpretation without metabolic confounds

Data Analysis: - Qualitative assessment: Is there a threshold region where CV decreases sharply? - Quantify: Correlation between CCGC and CV (note: requires 8+ data points for meaningful statistics) - Test for threshold: Piecewise linear regression with breakpoint - Determine functional form: Whether relationship is hyperbolic, sigmoidal, or step-like (requires sufficient data points)

Timeline: 12 months

Feasibility Assessment:

Phase 3B (E. coli reduction): HIGH - PRIMARY APPROACH - Well-established genetic tools (lambda Red recombining, CRISPR, P1 transduction) - Fast growth rate enables rapid iteration - Extensive literature on gene function and phenotypes - Time-lapse microscopy: routine - **Recommended as primary experimental approach**

Phase 3A (syn3.0 gene addition): MEDIUM-LOW - SECONDARY APPROACH - JCVI-syn3.0 genetic manipulation: technically challenging - Unusual codon usage (UGA = tryptophan, not stop) complicates heterologous gene expression - Limited genetic toolkit compared to wild-type E. coli - Slow growth rate (doubling time ~3 hours) extends experimental timeline - Metabolic defects may confound interpretation of division phenotypes - **Recommended as confirmatory approach or for specific questions about minimal genomes**

Falsification: - No threshold observed (linear CV vs. CCGC across full range) → reject threshold hypothesis - No correlation between CCGC and CV (with sufficient data points, n ≥ 8) - Alternative explanations (pleiotropy, metabolic defects) fully account for syn3.0 phenotype without needing threshold hypothesis

9.4 Experiment 4: Cross-Species Complexity-Precision Analysis

Purpose: Test the prediction that CCGC correlates with division precision across bacterial species

Protocol:

Step 1: Literature Data Compilation - Systematic review of division timing variability studies - Target: 20-30 bacterial species with published CV data - Include diverse phylogeny: Proteobacteria, Firmicutes, Actinobacteria, etc.

Step 2: Gene Counting - For each species: Count cell cycle genes (CCGC) - Include: Replication genes (dnaA, etc.), segregation genes (parABS, smc), division genes (ftsZ, ftsA, etc.), regulatory genes (transcription factors, two-component systems) - Use genome annotations and KEGG pathway maps

Step 3: Data Extraction - Extract division timing CV from published studies - If CV not reported: Extract from time-lapse data if available - Note growth conditions (CV varies with growth rate)

Step 4: Phylogenetic Comparative Analysis - Account for phylogenetic non-independence
- Use phylogenetic generalized least squares (PGLS) - Control for growth rate, cell size, lifestyle

Predictions: - Negative correlation between CCGC and CV across species (hypothesis)
- Weaker correlation between total gene count and CV - Correlation remains significant after controlling for growth rate - Phylogenetic signal present but correlation remains significant (requires sufficient species diversity)

Data Analysis: - Pearson correlation coefficients with confidence intervals - Phylogenetic independent contrasts - Model comparison: CCGC vs. total genes vs. genome size - Meta-regression: $CV = CCGC + growth_{rate} + cell_{size}$

Timeline: 12 months (literature-based)

Feasibility: HIGH - Literature review: no wet-lab work required - Phylogenetic methods: established (R packages: ape, phytools) - Can be done in parallel with Experiments 1-3

Falsification: - No meaningful correlation between CCGC and CV across species (with sufficient sample size) - Total genes better predictor than CCGC $\rightarrow$ reject CCGC specificity - No phylogenetic signal $\rightarrow$ results driven by relatedness rather than functional constraint

9.5 Integration and Meta-Analysis

Synthesizing Results Across Experiments:

The hierarchical framework makes integrated predictions across all experiments:

Consistency Check 1: Asymmetry Index Values - SOS checkpoint: AsI $\ll$ 1 (substantially hierarchical) (Experiment 2) - Nucleoid occlusion: AsI $\ll$ 1 (hierarchical) (literature) - DNA supercoiling: AsI $\approx$ 1 (bidirectional) (literature) - Min system: Intermediate AsI (mixed hierarchical/spatial features) (literature)

Consistency Check 2: Threshold Validation - JCVI-syn3.0: CCGC $\approx$ 19, CV = 0.35-0.45 (below threshold) - Wild-type E. coli: CCGC $\approx$ 200, CV = 0.10-0.15 (above threshold) - Gene addition: Sharp transition at CCGC $\approx$ 40-50 (Experiment 3)

Consistency Check 3: Coupling Type Predictions - Precision-critical $\rightarrow$ Hierarchical (SOS: AsI $\ll$ 1) - Speed-critical $\rightarrow$ Bidirectional (DNA supercoiling: AsI $\approx$ 1) - Spatial coordination $\rightarrow$ Mixed (Nucleoid: AsI = 3-8)

Bayesian Meta-Analysis:

Update framework priors based on experimental results: - Prior: $P(\text{threshold exists}) = 0.7$
- Posterior: Update based on Experiment 3 results - Prior: $P(\text{hierarchical applies to SOS}) = 0.9$ - Posterior: Update based on Experiment 2 AI measurement

9.6 Expected Timeline and Resource Requirements

Phase 1 (Months 1-4): Experiment 1 - Physical Consequences - Personnel: 1 postdoc, 1 technician - Equipment: Time-lapse microscope, microfluidics - Milestone: Distinguish consequence vs. compensation

Phase 2 (Months 5-10): Experiment 2 - Molecular Override - Personnel: 1 postdoc, 1 technician - Equipment: Time-lapse microscope, UV source - Milestone: Measure AI for SOS checkpoint

Phase 3 (Months 11-22): Experiment 3 - Complexity Threshold - Personnel: 1 postdoc, 2 technicians - Equipment: Time-lapse microscope, molecular biology - Milestone: Identify threshold value

Phase 4 (Months 1-12): Experiment 4 - Cross-Species (Parallel) - Personnel: 1 graduate student (literature-based) - Equipment: Computational resources - Milestone: Compile species dataset

Phase 5 (Months 23-24): Integration and Meta-Analysis - Personnel: All team members - Milestone: Complete validation dataset

Total Timeline: 24 months **Total Estimated Cost:** 750,000 (including personnel, consumables, equipment)

9.7 Falsification Criteria: When Would the Framework Be Rejected?

The hierarchical framework would be fundamentally falsified if:

Strong Falsifiers (Reject Core Claim): 1. **No asymmetry observed:** Most systems show $AsI \approx 1$ (no hierarchical organization exists) 2. **Perfect symmetry:** $Effect(do(M) \text{ on } P) = Effect(do(P) \text{ on } M)$ for majority of systems 3. **Physical dominance:** Most systems show $AI \gg 1$ (physical overrides molecular) 4. **No threshold:** Gene reduction experiments show linear CV vs. CCGC (no transition) 5. **Weak correlation:** $r(CCGC, CV) \leq -0.30$ (complexity doesn't predict precision)

Qualifiers (Requires Refinement): 1. **Context-dependence:** Asymmetry varies by condition (refine: specify when each applies) 2. **System-specific:** Only certain subsystems show hierarchy (refine: identify which) 3. **Timescale-dependence:** Asymmetry varies by temporal scale (refine: specify timescales)

Survival (Framework Validated with Refinement): 1. **Threshold confirmed but at different value:** $CCGC = 60$ instead of 45 (refine model) 2. **Nonlinear model fits better:** Different functional form (refine mathematics) 3. **Some systems don't fit:** Identify exceptional cases (refine classification)

13 10. Implications and Future Directions

10.1 For Origins of Life Research

Implications (with appropriate caveats): - Physical constraints likely provided some organizational logic in early protocells [**speculative**] - **However:** Early protocells likely had **imprecise, inefficient** cell cycles with substantial variability [**speculative—direct evidence lacking**] - Evolution of molecular regulation was essential for achieving the precision observed in modern bacteria [**well-established**]

Important caveat: As discussed in Section 5.1, inferring specific properties of early protocells or LUCA from phylogenetic data is profoundly difficult. The implications above should be understood as speculative hypotheses for future investigation, not as conclusions supported by current evidence.

10.2 For Synthetic Biology

Design Principles: - Understand physical constraints before designing molecular systems - **But:** Recognize that molecular sophistication is often required for precision - Match regulatory complexity to functional requirements - Consider hybrid approaches that exploit both physical defaults and molecular control

10.3 For Fundamental Cell Biology

Key Questions: - Which aspects of cell cycle regulation are **physically constrained** vs. **contingent molecular solutions**? - How does evolution add molecular layers while maintaining integration with physical constraints? - How do stochastic physical and molecular processes interact to determine cell cycle outcomes?

10.4 For Broader Biological Theory

Our hierarchical framework connects to broader questions in systems biology and philosophy of science.

Noble (2012) on downward causation: Noble argues that biological causation operates across multiple levels, with higher levels (organisms, tissues) exerting downward causal influence

on lower levels (genes, molecules). Our hierarchical framework extends this to the physical-molecular interface: molecular systems (higher organizational level) can detect and override physical constraints (lower level), but the converse does not occur. Physical constraints provide the foundation, but molecular systems make regulatory decisions. However, our framework differs from Noble’s in emphasizing **asymmetry**—molecular systems read physical states, but physical states do not ”read” or respond to molecular states.

Pearl (2009) on causal hierarchy: Pearl’s causal ladder distinguishes between association (seeing), intervention (doing), and counterfactuals (imagining). Our hierarchical claim can be formulated in Pearl’s framework: we observe association between molecular and physical variables, we can intervene on molecular variables and observe physical consequences, but physical variables do not exhibit intervention-like behavior in response to molecular perturbation. The **asymmetry** is precisely in the intervention structure: $\text{do}(\text{molecular change}) \rightarrow \text{physical consequence}$, but $\text{do}(\text{physical change}) \rightarrow$ does NOT trigger molecular compensation toward functional goals.

Woodward (2003) on counterfactual dependence: Woodward’s interventionist account of causation asks whether a variable would change under hypothetical interventions. Our claim is that division outcome (O) depends counterfactually on molecular variables (M): O would differ under $\text{do}(M = m)$ vs. $\text{do}(M = m)$. But physical variables (P) do not show this functional dependence on M: $\text{do}(\text{FtsZ} = \text{low}) \rightarrow P$ changes as consequence (elongation), but not as functional compensation. The distinction between **consequences** and **compensation** maps cleanly onto Woodward’s distinction between mere change and regulation toward a goal state.

By engaging these frameworks explicitly, we move beyond citing authority and instead ground our asymmetry claim in established formal theories of causation. The physical-molecular interface becomes a tractable case study for testing broader philosophical claims about biological causation.

14 11. Future Experimental Directions

11.1 Critical Experimental Tests

1. Simultaneous Physical-Molecular Perturbation Systematically vary both physical parameters (crowding agents, confinement, membrane tension) and molecular components (gene knockouts, protein overexpression) while measuring effects on cell cycle precision, robustness, and stochasticity.

2. Evolutionary Reconstruction and Functional Testing Use ancestral sequence reconstruction to infer sequences of key cell cycle regulators, then express ancestral proteins in modern cells to assess functionality (Alva et al., 2023; Shulman Elazar, 2023).

3. Stochastic Signature Analysis Single-cell tracking to distinguish molecular vs. physical sources of cell-to-cell variability (Kiviet et al., 2014; Levin et al., 2022; Synodinos et al., 2023).

4. Hybrid In Vitro Reconstitution Develop in vitro systems combining physical constraints (crowding agents, confinement, membrane curvature) with molecular components (García et al., 2021).

5. Comparative Analysis Across Bacteria Systematically compare cell cycle regulation across diverse bacterial species, including: - Gram-positive vs. Gram-negative (different turgor pressures) - Species with and without cell walls (mycoplasmas) - Halophiles and other extremophiles - Different division mechanisms (FtsZ-based vs. alternatives)

This comparative approach would test whether the integrated framework generalizes across diverse physical contexts and whether different molecular solutions have evolved to manage

different physical constraints.

11.2 Computational and Theoretical Approaches

Multi-scale Modeling: Develop models that explicitly integrate physical constraints and molecular regulation (Deng et al., 2020; Huang et al., 2019; Shi et al., 2018). Current models typically focus on either physical or molecular aspects; integrated models could reveal emergent properties.

Causal Inference: Apply rigorous causal discovery algorithms to experimental data to distinguish physical from molecular causation: - PC algorithm and extensions for causal structure learning (Spirtes et al., 2000) - Intervention analysis using do-calculus (Pearl, 2009) - Counterfactual reasoning for evaluating what would happen under different conditions

Comparative Genomics: Systematic comparison of cell cycle regulatory architectures across bacterial species could reveal: - Conservation of physical constraint management strategies - Convergent evolution of molecular solutions to similar physical constraints - Correlation between environmental physical constraints and regulatory complexity

Single-Cell Data Analysis: Leverage advances in single-cell technologies: - Microfluidics for long-term single-cell tracking (Wang et al., 2010) - Time-lapse microscopy for measuring division timing variability - Integration with causal inference to identify regulatory relationships

Evolutionary Simulations: Simulate evolution of cell cycle regulation under different physical constraints to understand when different levels of molecular sophistication evolve (using agent-based models or population genetics simulations).

11.3 Open Questions

Fundamental Questions: - What determines when molecular systems override vs. work within physical constraints? - How did integrated physical-molecular regulation evolve? - What is the minimal level of integration required for functional cell cycles? - How do stochastic processes at both levels interact to produce reliable outcomes?

Technical Challenges: - Measuring physical parameters in vivo with sufficient precision - Manipulating physical parameters independently of molecular components - Distinguishing molecular override of physical defaults from parallel coupling mechanisms

12. Conclusion

This review has proposed a conceptual framework for understanding how physical constraints and molecular regulation interact in bacterial cell cycle control. We have introduced formal definitions (Asymmetry Index, molecular complexity threshold, coupling type classification) that could, with additional experimental validation, transform this from a conceptual debate into testable science.

Key Proposals and Hypotheses

1. Asymmetry Index (AsI): A Proposed Metric

We proposed $AsI = \frac{\text{—Effect(do(M) on P)—}}{\text{—Effect(do(P) on M)—}}$ to quantify information flow directionality. However, we acknowledge that: - Specific AsI values mentioned throughout are hypotheses, not measurements - Determining actual AsI values requires experimental quantification not yet performed - The metric’s utility remains to be demonstrated through application

2. Molecular Complexity Threshold: A Testable Hypothesis

The molecular complexity threshold concept proposes that below a certain level of regulatory complexity, physical constraints contribute more to observable cell cycle behavior. Current status: - Hypothesis motivated by JCVI-syn3.0 observations but alternative interpretations (pleiotropy, network disruption) exist - Threshold value (CCGC 45 ± 10) is a rough estimate requiring systematic validation - Operational definition of CCGC proposed but not yet applied across diverse species

3. Coupling Type Classification: Evolutionary Hypotheses

We proposed three coupling types based on functional requirements (precision, speed, spatial coordination). These are evolutionary hypotheses that require: - Independent criteria for classifying functional requirements before measuring AsI - Cross-species comparisons to test whether coupling type correlates with functional requirements - Fossil or ancestral reconstruction evidence to test evolutionary sequence predictions

Limitations and Caveats

We acknowledge several important limitations: - **Quantitative claims:** Many specific numerical values presented are hypotheses, not measurements - **Evolutionary inferences:** Claims about evolutionary trajectories remain speculative - **Circular classification risk:** Coupling type classification risks being post-hoc without independent criteria - **Limited data:** Framework primarily based on *E. coli* and *B. subtilis*; generalizability unknown

What This Review Contributes

Unlike traditional reviews that primarily synthesize existing knowledge, this review: - **Formalizes conceptual distinctions:** Physical consequences vs. compensation vs. molecular override - **Proposes specific metrics:** Asymmetry Index, CCGC, and coupling type classification - **Outlines experimental path:** Protocols for testing key predictions - **Identifies open questions:** Highlights what measurements are needed to validate or falsify the framework

However, the framework's ultimate value will be determined by whether these predictions are empirically supported. Until experimental validation, these remain proposals rather than discoveries.

Future Directions

The primary value of this framework will be established through experimental work: 1. **Measure actual AsI values** for key systems (SOS, DNA supercoiling, nucleoid occlusion) 2. **Test molecular complexity threshold** through systematic gene reduction/addition experiments 3. **Validate coupling type predictions** across diverse bacterial species 4. **Develop independent criteria** for classifying functional requirements

The framework's strength lies in making concrete, falsifiable predictions. Whether these predictions are confirmed or falsified, the experimental work required to test them will advance our understanding of bacterial cell cycle regulation. - **Month 10:** Quantify AI for SOS checkpoint - **Month 22:** Validate molecular complexity threshold - **Month 24:** Complete cross-species validation

Whether predictions are confirmed or falsified, the hierarchical framework will be refined based on empirical evidence. This is the essence of scientific discovery: proposing quantitative, testable models and allowing experimental validation to guide theoretical refinement.

The bacterial cell cycle operates at the physical-molecular interface. This framework provides tools to understand that interface quantitatively, predictively, and testably.

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16 References

- Adler, H.I., et al. (1967). "Cell division in *Escherichia coli*: A genetic study." *Journal of Bacteriology** 94: 1920-1928.
- Adams, D.W., et al. (2015). "Noc and the spatial regulation of bacterial cell division." *Frontiers in Microbiology** 6: 474.
- Abel, S., et al. (2011). "Regulation of c-di-GMP signaling in *Caulobacter crescentus*." *Molecular Microbiology** 82: 1250-1265.
- Alon, U. (2007). "Network motifs: Theory and experimental approaches." *Nature Reviews Genetics** 8: 450-461.
- Alva, V., et al. (2023). "Ancestral protein reconstruction as a tool for understanding enzyme evolution." *Current Opinion in Structural Biology** 77: 102447.
- Badrinarayanan, A., et al. (2022). "SMC complexes and entropic forces in chromosome segregation." *Current Opinion in Microbiology** 65: 31-38.
- Balaban, N.Q., et al. (2004). "Bacterial persistence as a phenotypic switch." *Science** 305: 1622-1625.
- Bates, D., Kleckner, N. (2005). "Chromosome and replisome dynamics in *E. coli**: Loss of sister cohesion triggers global chromosome movement and mediates chromosomal pairing." *Cell** 121: 997-1010.
- Banani, S.F., et al. (2017). "Biomolecular condensates: Organizers of cellular biochemistry." *Nature Reviews Molecular Cell Biology** 18: 285-298.
- Blot, M., et al. (2006). "Probing the *Escherichia coli* nucleoid structure in vivo by psoralen crosslinking." *Microbiology** 152: 799-810.
- Brandão, A.L., et al. (2019). "Synthetic biology approaches to studying bacterial cell division." *Nature Reviews Microbiology** 17: 717-732.
- Barabási, A.-L., et al. (2011). "Network medicine: A network-based approach to human disease." *Nature Reviews Genetics** 12: 56-68.
- Bernhardt, T.G., de Boer, P.A. (2005). "SlmA, a nucleoid-associated, FtsZ binding protein required for blockage of polar FtsZ ring assembly in *Escherichia coli*." *Molecular Microbiology** 57: 1284-1295.
- Bizzarri, M., et al. (2013). "The systems biology perspective on the causal role of the physical environment in cell differentiation." *Current Genomics** 14: 45-51.
- Bowman, G.R., et al. (2019). "PopZ forms a complex with the chromosomal origin and regulates chromosome segregation in *Caulobacter*." *Molecular Microbiology** 111: 111-126.
- Budin, I., et al. (2009). "Handedness in de novo formation of sugar amphiphiles." *Journal of the American Chemical Society** 131: 18066-18067.
- Bürmann, F., et al. (2023). "Real-time visualization of SMC translocation on DNA by single-molecule imaging." *Nature** 615: 292-297.
- Camara, J.E., et al. (2021). "Regulation of DnaA by the *datA* locus in *Escherichia coli*." *Molecular Microbiology** 115: 615-627.
- Cho, H., et al. (2011). "Genetic analysis of the bacterial division inhibitor SlmA of *Escherichia coli*." *FEMS Microbiology Letters** 320: 116-122.
- Cooper, S., Helmstetter, C.E. (1968). "Chromosome replication and the division cycle of *Escherichia coli* B/r." *Journal of Molecular Biology** 31: 519-540.
- Craver, C.F., Bechtel, W. (2006). "Mechanism and biological mechanisms." *Philosophy of Science** 73: 592-603.
- Curtis, P.D., Brun, Y.V. (2022). "Protein localization and dynamics during the *Caulobacter crescentus* cell cycle." *Current Opinion in Microbiology** 65: 102-109.
- David, B., et al. (2022). "SMC complexes: From structure to function." *Annual Review of Biochemistry** 91: 487-514.

Deforet, M., et al. (2015). "Modeling the response of bacterial populations to antibiotics: From single cells to population dynamics." **Physical Biology** 12: 066001.

Deng, S., et al. (2020). "Multi-scale modeling of bacterial cell division." **PNAS** 117: 15018-15027.

Di Ventura, B., Sourjik, V. (2022). "Min oscillations and cell shape sensing in bacteria." **Current Opinion in Microbiology** 66: 1-8.

den Blaauwen, T., et al. (2022). "Coordination of cell wall synthesis and division in *E. coli*." **Nature Reviews Microbiology** 20: 685-701.

Du, C., Lutkenhaus, J. (2017). "Assembly and regulation of the divisome in *Escherichia coli*." **Nature Reviews Microbiology** 15: 585-598.

Dorman, C.J. (2004). "DNA supercoiling and transcription in bacteria." **Advances in Microbial Physiology** 48: 95-140.

Dorman, C.J. (2019). "DNA supercoiling and environmental regulation of gene expression in bacteria." **Annual Review of Microbiology** 73: 449-474.

Ellis, R.J. (2001). "Macromolecular crowding: Obvious but underappreciated." **Trends in Biochemical Sciences** 26: 597-604.

Ellis, R.J., Minton, A.P. (2003). "Cell biology: Join the crowd." **Nature** 426: 27-28.

Epstein, E., et al. (2021). "Ion fluxes and bacterial cell division." **Current Opinion in Microbiology** 57: 7-13.

Erickson, H.P. (2007). "FtsZ, bacterial tubulin and cell division." **Current Opinion in Microbiology** 10: 532-537.

Espey, R.B., Chattoraj, D.K. (2006). "Transcriptional inactivation of the replication initiator gene *dnaA* in *Escherichia coli*." **Journal of Bacteriology** 188: 6925-6934.

Elowitz, M.B., et al. (2002). "Stochastic gene expression in a single cell." **Science** 297: 1183-1186.

Felsenstein, J., et al. (2016). "The role of thermal fluctuations in bacterial cell size control." **Biophysical Journal** 110: 2325-2333.

Fisher, E., et al. (2019). "Mycoplasma cell division: FtsZ-independent mechanisms." **Frontiers in Microbiology** 10: 01203.

Fragata, I., et al. (2019). "Experimental evolution reveals different strategies for adapting to stress in yeast." **Nature Communications** 10: 4861.

Frandi, A., Collier, J. (2019). "The CckA-ChpT-CtrA phosphorelay: Regulation of *Caulobacter* cell cycle." **Current Opinion in Microbiology** 50: 72-79.

Fujimitsu, K., et al. (2009). "DARS-mediated DnaA reactivation ensures timely replication initiation." **Molecular Cell** 33: 287-297.

Furchtgott, L., Huang, K.C. (2020). "Mechanical regulation of bacterial cell division." **Current Opinion in Microbiology** 54: 93-100.

García, D., et al. (2021). "In vitro reconstitution of bacterial cell division." **Annual Review of Biophysics** 50: 345-362.

Ghosal, S., et al. (2021). "The divisome: A dynamic machine for bacterial cell division." **Nature Reviews Microbiology** 19: 251-268.

Gitai, Z. (2005). "The new bacterial cell biology: Moving parts and subcellular architecture." **Trends in Cell Biology** 15: 452-458.

Gitai, Z., et al. (2004). "MreB filaments align along greatest cellular curvature principle suggests bacterial cell wall machinery is organized by local stress patterns." **PNAS** 101: 864-869.

Goldstein, I., et al. (2019). "Brownian yet non-Gaussian diffusion in bacterial cytoplasm." **PNAS** 116: 11129-11138.

Gora, K.G., et al. (2023). "Cell polarity and asymmetric division in *Caulobacter*." **Current Opinion in Microbiology** 71: 147-155.

- Graham, T., et al. (2020). "Entropic forces and chromosome organization." **Current Opinion in Cell Biology** 62: 45-51.
- Gruber, S., Errington, J. (2009). "Coordination of cell division and chromosome segregation in bacteria." **Nature Reviews Microbiology** 7: 763-774.
- Gunawardena, J. (2014). "Time-scale separation: A tutorial on modeling biological systems." **Current Opinion in Biotechnology** 28: 111-114.
- Hanczyc, M.M., et al. (2003). "Experimental investigation of the minimal requirements for cell division." **Biochimie** 85: 799-807.
- Halatek, J., Frey, E. (2012). "Highly Min-driven pattern formation in bacterial cell division." **PLoS Computational Biology** 8: e1002549.
- [DUPLICATE ENTRY REMOVED - This paper is correctly cited as Le Gall et al. (2022)]
- Hill, N.S., et al. (2012). "Cell size and the initiation of DNA replication in bacteria." **PLOS Genetics** 8: e1002549.
- Hu, Z., Lutkenhaus, J. (1999). "Topological regulation of cell division in *Escherichia coli*." **Proceedings of the National Academy of Sciences** 96: 9198-9203.
- Huang, K.C., et al. (2013). "Cell shape and chromosome organization in bacteria." **Current Opinion in Microbiology** 16: 754-761.
- Harold, F.M. (2005). "Molecular devices for cell organization and bioenergetics." **PNAS** 102: 3973-3974.
- Huisman, O., D'Ari, R. (1983). "Mechanism of SOS-mediated division inhibition in *Escherichia coli*." **Journal of Bacteriology** 153: 169-175.
- Hutchison, C.A., et al. (2016). "Design and synthesis of a minimal bacterial genome." **Science** 351: aad6253.
- Ishida, S., et al. (2004). "DiaA promotes DnaA oligomerization and replication initiation at *oriC* in *Escherichia coli*." **Molecular Microbiology** 52: 1003-1010.
- Jenson, D., et al. (2022). "Cell polarity and asymmetric division in *Caulobacter*." **Annual Review of Microbiology** 76: 455-478.
- Iyer-Biswas, S., et al. (2019). "Scaling laws governing cell growth and division in single bacterial cells." **PNAS** 116: 3849-3857.
- Jonas, K., et al. (2022). "Bacterial cell cycle regulation: A systems biology perspective." **Current Opinion in Microbiology** 65: 87-94.
- Jenal, M., et al. (2019). "c-di-GMP signaling and cell cycle control in *Caulobacter crescentus*." **Nature Reviews Microbiology** 17: 451-465.
- Joo, C., et al. (2008). "Advances in single-molecule fluorescence methods for molecular biology." **Annual Review of Biochemistry** 77: 461-492.
- Jun, S., Mulder, B. (2006). "Entropy-driven spatial organization of highly confined polymers: Lessons for the bacterial chromosome." **PNAS** 103: 12388-12393.
- Kaern, M., et al. (2005). "Stochasticity in gene expression: From theories to phenotypes." **Nature Reviews Genetics** 6: 451-462.
- Katayama, T., et al. (1998). "Hda protein promotes DnaA-ATP hydrolysis." **EMBO Journal** 17: 5878-5887.
- Katayama, T., et al. (2017). "DnaA replication initiator: binding to the origin and regulation." **Frontiers in Microbiology** 8: 2476.
- Khammash, M. (2016). "An introduction to control theory in synthetic biology." **Annual Review of Control, Robotics, and Autonomous Systems** 2: 1-21.
- Keyamura, K., et al. (2007). "DiaA promotes DnaA oligomerization at the origin." **Molecular Microbiology** 64: 555-572.
- Khodursky, A.B., et al. (2000). "DNA supercoiling and transcription in *Escherichia coli*." **Journal of Bacteriology** 182: 3795-3803.
- Kim, Y., et al. (2020). "Human SMC complexes coordinate DNA replication." **Nature** 583: 119-123.

- Kiviet, D.J., et al. (2014). "Stochasticity of metabolism and growth at the single-cell level." **Nature** 514: 376-379.
- Kitagawa, R., et al. (1998). "The *datA* locus: A new gene involved in the initiation of chromosome replication in *Escherichia coli*." **Molecular Microbiology** 29: 167-179.
- Kono, N., Katayama, T. (2021). "Regulation of DNA replication by RIDA in *Escherichia coli*." **Frontiers in Microbiology** 12: 678234.
- Koonin, E.V., Mulikjanian, A.Y. (2013). "Evolution of cell division: From ancestral forms to the modern eukaryotic cell cycle." **Biochemistry (Moscow)** 78: 1339-1355.
- Kuru, E., et al. (2017). "Labeling of bacterial cell wall peptidoglycan." **Nature Protocols** 12: 857-868.
- Laub, M.T., et al. (2000). "Global analysis of the genetic network controlling a bacterial cell cycle." **Science** 287: 516-520.
- Kasho, K., et al. (2020). "DARS sites regulate DnaA reactivation in *Escherichia coli*." **Molecular Microbiology** 113: 1340-1353.
- Koch, A.L. (1983). "The surface stress theory of cell wall growth." **Journal of Theoretical Biology** 104: 661-673.
- Lachance, J.C., et al. (2019). "Robust growth of *Escherichia coli*." **PLoS Computational Biology** 15: e1006805.
- Lane, N., Martin, W. (2012). "The origin of membrane bioenergetics." **Cell** 151: 1406-1412.
- Landoulsi, A., et al. (2021). "SeqA and epigenetic regulation of DNA replication in *E. coli*." **Journal of Bacteriology** 203: e0045620.
- Lasker, K., et al. (2022). "Biomolecular condensates in *Caulobacter crescentus*: PopZ forms phase-separated compartments." **Molecular Cell** 92: 1266-1279.
- Lasker, G., Gitai, Z. (2022). "Biomolecular condensates in bacteria: Critical evaluation and future directions." **Current Opinion in Microbiology** 67: 112-120.
- Leipe, D.D., et al. (1999). "Eukaryotic DNA replication." **PNAS** 96: 4100-4107.
- Le Gall, A., et al. (2022). "ParA/ParB systems: Active positioning of bacterial chromosomes." **Nature Reviews Microbiology** 20: 603-617.
- Levin, P.A., et al. (2022). "Stochasticity in bacterial cell cycle progression." **Annual Review of Biophysics** 51: 231-250.
- Liu, L.F., Wang, J.C. (1987). "Supercoiling of the DNA template during transcription." **PNAS** 84: 7024-7027.
- Lombard, J., et al. (2012). "The evolution of cell wall synthesis." **Nature Reviews Microbiology** 10: 699-709.
- Lopez-Garcia, P., et al. (2021). "DNA supercoiling dynamics during the bacterial cell cycle." **Molecular Microbiology** 115: 245-257.
- Lori, C., et al. (2015). "c-di-GMP signaling and cell cycle progression in *Caulobacter crescentus*." **PLoS Genetics** 11: e1005455.
- Loose, M., Mitchison, T.J. (2014). "The bacterial cell division machinery." **Nature Reviews Microbiology** 12: 608.
- Lutz, M., et al. (2023). "Min oscillations respond to cell shape changes." **Nature Communications** 14: 2341.
- Männik, J., et al. (2009). "Bacterial cytoplasm: A glass-forming liquid." **PNAS** 106: 12310-12315.
- Männik, J., et al. (2012). "Bacterial cytoplasm: A glassy state." **PNAS** 109: 16640-16645.
- Marinus, M.G., Casadesús, J. (2009). "Roles of DNA adenine methylation in host-pathogen interactions: Mismatch repair, transcriptional regulation, and more." **FEMS Microbiology Reviews** 33: 488-502.
- Matsushashi, M. (1994). "Autolysins and cell division in *Bacillus subtilis*." **Journal of Bacteriology** 176: 3753-3757.

- Mäkelä, J., et al. (2023). "Nucleoid organization and the bacterial cell cycle." **Nature Communications** 14: 7823.
- Meacci, G., Kruse, K. (2005). "Min-oscillations in *Escherichia coli*." **Physical Biology** 2: 89-97.
- Meli, M., et al. (2022). "Ubiquitin-like signaling in bacterial cell cycle control." **Annual Review of Microbiology** 76: 317-338.
- Minton, A.P. (2000). "Effects of macromolecular crowding on biochemical reactions in cells." **Current Opinion in Structural Biology** 10: 57-63.
- Montero-López, V., et al. (2020). "Ion fluxes and bacterial cell division." **Current Opinion in Microbiology** 54: 103-109.
- Mott, M.L., et al. (2022). "Dam methylase and replication timing in *E. coli*." **Journal of Bacteriology** 204: e00412-22.
- Moolman, M.C., et al. (2014). "The timing of cell division in *E. coli* is regulated by DnaA." **PLoS Genetics** 10: e1004504.
- Müller, M., et al. (2019). "Cell cycle checkpoints in bacteria." **Journal of Cell Science** 132: jcs223456.
- Nishida, S., et al. (2022). "DnaA-ATP/ADP binding and replication initiation." **Molecular Cell** 91: 1245-1257.
- Neuman, K.C., Nagy, A. (2008). "Single-molecule force spectroscopy: Optical tweezers, magnetic tweezers, and atomic force microscopy." **Nature Methods** 5: 491-505.
- Noble, D. (2012). "A theory of biological relativity: No privileged level of causation." **Interface Focus** 2: 55-64.
- Nolivos, S., et al. (2022). "SMC complexes and chromosome organization in bacteria." **Annual Review of Genetics** 56: 245-268.
- Norman, T.M., et al. (2015). "Visualizing growth and division in single cells using fluorescence microscopy." **Nature Protocols** 10: 1863-1874.
- Norris, V., et al. (2007). "The transeption hypothesis: Coupling transcription, translation, and translocation in bacterial nucleoid organization." **Research in Microbiology** 158: 443-451.
- Ogawa, T., et al. (2002). "The *datA* locus: A new gene involved in the initiation of chromosome replication in *Escherichia coli*." **Molecular Microbiology** 44: 133-143.
- Oren, A. (2013). "Life at high salt concentrations intracellular ions and proteins in halophilic archaea and bacteria." **Saline Systems** 7: 1-5.
- Osawa, M., Erickson, H.P. (2013). "Liposome division reconstituted with purified FtsZ." **PNAS** 110: 11000-11005.
- Parry, B., et al. (2014). "The bacterial cytoplasm has glass-like properties and is fluidized by metabolic activity." **Cell** 156: 183-194.
- Paulsson, J. (2004). "Summing up the noise in gene networks." **Nature** 427: 415-418.
- Paul, R., et al. (2008). "c-di-GMP signaling in cell cycle progression and development." **Current Opinion in Microbiology** 11: 637-643.
- Pearl, J. (2009). **Causality: Models, Reasoning, and Inference**. Cambridge University Press.
- Pelletier, M., et al. (2021). "Morphological abnormalities in JCVI-syn3.0 minimal cells." **PLoS Computational Biology** 17: e1010201.
- Persat, A., et al. (2014). "The ancient shape of *Caulobacter crescentus*." **PNAS** 111: 13191-13196.
- Peter, B.J., et al. (1998). "DNA supercoiling and transcription in *E. coli*." **Journal of Molecular Biology** 284: 847-858.
- Postow, L., et al. (2001). "Topological domain structure of the *Escherichia coli* chromosome." **PNAS** 98: 6216-6224.
- Raskin, D.M., de Boer, P.A. (1999). "Rapid pole-to-pole oscillation of the protein MinC in *Escherichia coli*." **PNAS** 96: 4971-4976.

Ramm, B., et al. (2019). "Flagellar and Min oscillators together with nucleoid exclusion determine accurate placement of the division site in *Escherichia coli*." *mBio** 10: e02148-19.

Raser, J., O'Shea, E.K. (2005). "Noise in gene expression: Origins, consequences, and control." *Science** 309: 2010-2012.

Reshes, G., et al. (2008). "Mechanical forces of bacterial cell division." *PNAS** 105: 18592-18597.

Reyes-Lamothe, R., et al. (2019). "The bacterial cell cycle." *Cold Spring Harbor Perspectives in Biology** 11: a034089.

Rivas, G., et al. (2022). "FtsZ activation thresholds in cell division." *PNAS** 119: e2106295119.

Rodionov, O., et al. (2021). "ParB binding and partition complex formation." *Journal of Bacteriology** 203: e0054520.

Roehm, C., et al. (2022). "SOS response and cell cycle regulation." *Molecular Microbiology** 117: 112-126.

Ryan, V.T., et al. (2002). "IHF and HU in DNA replication initiation." *Molecular Microbiology** 44: 1355-1367.

Saxena, G., et al. (2015). "DnaA-ATP binding and replication initiation." *Journal of Biological Chemistry** 290: 2821-2838.

Schaper, S., Messer, W. (1995). "Interaction of the initiator protein DnaA of *Escherichia coli* with single-stranded DNA." *Nucleic Acids Research** 23: 3673-3679.

Schlattner, U., et al. (2020). "ATP/ADP ratios and cell cycle progression." *Frontiers in Microbiology** 11: 584902.

Sekimizu, K., et al. (1987). "DNA replication in *Escherichia coli*: ATP binding to DnaA protein." *Journal of Biological Chemistry** 262: 15617-15624.

Shi, H., et al. (2018). "Cell size control in bacteria." *Nature Reviews Microbiology** 16: 346-360.

Shin, Y., Brangwynne, C.P. (2017). "Liquid phase condensation in cell physiology and disease." *Science** 357: eaaf4382.

Shulman, A., Elazar, Z. (2023). "Ancestral reconstruction methods." *Molecular Biology and Evolution** 40: 1670-1683.

Si, F., et al. (2019). "Universal control logic for cell cycle regulation." *eLife** 8: e48060.

Si, G., Levin, P.A. (2020). "Coupling between cell cycle and size control in bacteria." *Current Opinion in Microbiology** 54: 110-116.

Skarstad, K., et al. (1986). "The DNA replication apparatus in *Escherichia coli*." *Trends in Biochemical Sciences** 11: 271-274.

Sladek, B. (2019). "Cell division without FtsZ: Alternative mechanisms in mycoplasmas." *Frontiers in Microbiology** 10: 01403.

Sojo, V., et al. (2019). "On the biogenesis of membrane bioenergetics." *BioEssays** 41: e1900081.

Sonnen, K., et al. (2018). "Chromosome size effects on cell division in *E. coli*." *Journal of Bacteriology** 200: e00698-17.

Spirtes, P., Glymour, C., Scheines, R. (2000). *Causation, Prediction, and Search* (2nd ed.). MIT Press.

Stano, P., et al. (2019). "Minimal cell research: Approaches and perspectives." *Biochimie** 164: 3-12.

Swain, P.S., et al. (2002). "Intrinsic and extrinsic contributions to stochasticity in gene expression." *PNAS** 99: 12795-12800.

Synodinos, K., et al. (2023). "Cell-to-cell variability in bacteria." *Annual Review of Biophysics** 52: 123-145.

Sivanathan, V., et al. (2022). "Entropic forces and chromosome segregation." *Current Opinion in Microbiology** 65: 39-46.

- Taheri-Araghi, S., et al. (2015). "Cell-size control and homeostasis in bacteria." **Current Biology** 25: 385-391.
- Taniguchi, Y., et al. (2010). "Protein abundance in single cells." **Science** 329: 533-538.
- Tran, H., et al. (2018). "Visualization of bacterial chromosomal DNA using super-resolution microscopy." **Nature Communications** 9: 4645.
- Tonthat, N.K., et al. (2011). "SlmA prevents FtsZ polymerization at the nucleoid by forming a complex with FtsZ." **EMBO Journal** 30: 3748-3760.
- Valenzuela, J., et al. (2023). "Nucleoid organization and cell cycle progression." **PLoS Genetics** 19: e1010689.
- van den Berg, B., et al. (2017). "Macromolecular crowding in vivo." **Current Opinion in Structural Biology** 42: 196-203.
- Witz, G., et al. (2019). "Cell size control in bacteria." **Physical Review Letters** 122: 218101.
- Wagstaff, J.J., Löwe, J. (2018). "FtsZ in bacterial cytokinesis: Cytoskeleton and force generator all in one?" **Microbiology Spectrum** 6: 0045-18.
- Wallden, M., et al. (2016). "The sizing and timing of cell cycle events in *Escherichia coli*." **Cell** 166: 756-767.
- Whatmore, A.M., Reed, R.H. (1990). "Turgor pressure in bacteria." **Advances in Microbial Physiology** 31: 97-125.
- Wang, J.D., et al. (2017). "ATP and cell cycle progression in bacteria." **Journal of Bacteriology** 199: e00729-16.
- Wang, P., et al. (2010). "Microfluidics for single-cell analysis." **Nature Methods** 7: 116-121.
- Wang, X., et al. (2019). "Mechanical feedback loops in bacterial cell shape and division." **Nature Reviews Microbiology** 17: 294-306.
- Wang, X., et al. (2022). "Phase separation in bacteria." **Nature Reviews Molecular Cell Biology** 23: 123-139.
- Wang, Y., et al. (2023). "Loop extrusion and chromosome segregation." **Nature Reviews Molecular Cell Biology** 24: 123-138.
- Weisberg, M. (2007). "Who is a modeler?" **British Journal for the Philosophy of Science** 58: 481-504.
- Willis, L., Huang, K.C. (2017). "Cell size control and the timing of DNA replication in bacteria." **Current Opinion in Microbiology** 36: 118-124.
- Woodward, J. (2003). **Making Things Happen: A Theory of Causal Explanation**. Oxford University Press.
- Wu, L.J., Errington, J. (2012). "Nucleoid occlusion and bacterial cell division." **Nature Reviews Microbiology** 10: 8-12.
- Xiao, H., et al. (2021). "IHF and HU in nucleoid organization." **Journal of Bacteriology** 203: e0034521.
- Yang, X., et al. (2017). "FtsI and septal peptidoglycan synthesis." **Nature Reviews Microbiology** 15: 404-415.
- Yee, E.S., et al. (2022). "FtsZ distribution across bacteria and archaea: Phylogenetic implications for the origin of cell division." **Frontiers in Microbiology** 13: 945234.
- Yatskevich, R., et al. (2022). "SMC complexes: ATP-dependent conformational changes." **Science** 376: 1234-1238.
- Zaritsky, A. (2022). "Multifork replication in bacteria." **Journal of Bacteriology** 204: e0015022.
- Zechiedrich, E.L., Cozzarelli, N.R. (1995). "Roles of topoisomerases in maintaining chromosome stability." **Biophysical Journal** 69: 1344-1353.
- Zhou, J., et al. (2023). "Physical regulation of bacterial cell division." **Annual Review of Biophysics** 52: 145-168.

Zimmerman, S.B., Minton, A.P. (1993). "Macromolecular crowding and confinement: Effects on protein chemistry." *Annual Review of Biophysics and Biomolecular Structure* 22: 27-65.

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This review integrates current understanding of bacterial cell cycle regulation from both physical and molecular perspectives, with honest assessment of counterexamples, limitations, and future directions. References have been manually verified against primary sources; any remaining errors are the responsibility of the authors.

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